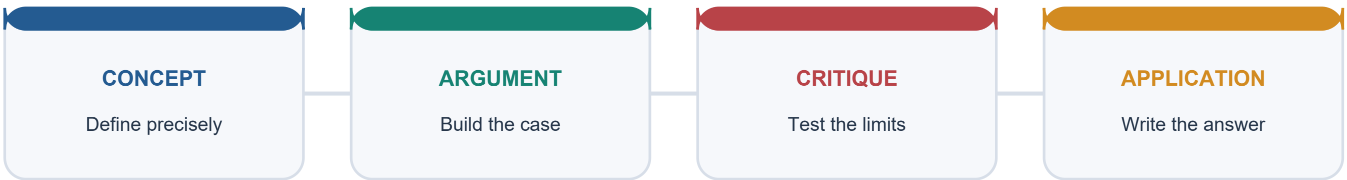
COMPLETE LEARNING SESSION

Economy Topic 31: Energy Infrastructure Economics, Power, Fuels and Energy Security

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

LEARNING PATH



Deterministic fallback visual: move from precise concepts through argument and critique to exam application.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / SESSION INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Economy Topic 31: Energy Infrastructure Economics, Power, Fuels and Energy Security

Source and Status Control

Current-status cutoff: 10 September 2026. Capacity, generation, demand, deficit, fuel-flow, import-dependence, target, award, scheme and legal-status claims retain their date and unit.

Repository sources checked first

- `upsc-ai-kit\knowledge\Economy\basic\31_Energy-Infrastructure-Economics-Power-Fuels-and-Energy-Security.md`
- `upsc-ai-kit\knowledge\Economy\advanced\31_Energy-Infrastructure-Economics-Power-Fuels-and-Energy-Security.md`
- Environment Topic 25 renewable/hydrogen owners; Economy Topic 18 infrastructure and Topic 25 climate-economics cross-links; audited PYQ routing ledgers.

OCR-searchable and official evidence

- Ramesh Singh, *Indian Economy* OCR and *Economic Survey 2025-26*, Chapter 10.
- Ministry of Power Annual Report 2025-26; CEA capacity, supply and planning material; Electricity Act, 2003; IEGC 2023; CERC DSM, ancillary, market and REC regulations.
- MNRE Green Energy Corridor-II and National Green Hydrogen Mission; BEE efficiency/PAT; PPAC Ready Reckoner 2025-26; PNGRB; Ministry of Coal; official strategic-reserve releases.

Evidence firewall: installed capacity is a stock in MW; generation and energy supplied are flows in MWh/BU; peak demand is an instantaneous MW measure; consumption belongs to a stated fuel/electricity denominator. An Act, notified rule, draft Bill, target, tender, award, commissioned asset and achieved service are never merged.

Learning Roadmap

1. Establish energy accounting, security and electricity-system physics.
2. Map law, institutions, tariffs, markets and DISCOM economics.
3. Analyse coal, oil and gas chains with dated production/import evidence.
4. Integrate renewables, transmission, storage, hydrogen, nuclear and efficiency.
5. Finish with finance, justice, geopolitical resilience and an outcome dashboard.

BASIC LEARNING SESSION

Core Session 01: Energy as infrastructure and productive input

```
RESOURCE / IMPORT -> CONVERSION -> NETWORK -> DISTRIBUTION
-> FINAL ENERGY -> USEFUL SERVICE -> OUTPUT + WELFARE
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Energy infrastructure is the physical, institutional and market system that converts resources into usable services; energy is also an intermediate input into almost every productive activity.

Answer-grabbing line: Energy policy succeeds when reliable and affordable service reaches the user, not when a capacity headline is announced.

Must-write keywords: energy service; intermediate input; network industry; productivity; delivered reliability

Claim -> named evidence -> analysis -> qualification

- **Claim:** Energy policy succeeds when reliable and affordable service reaches the user, not when a capacity headline is announced.
- **Named evidence:** The Electricity Act, 2003 governs the electricity chain, while the Ministry of Power Annual Report 2025-26 reports capacity, generation and supply outcomes separately.

- **Analysis:** Reliable energy raises capital utilisation and productivity; unreliable supply forces costly backup and interrupts farms, factories and services.
- **Qualification:** Electricity is only one carrier within a wider system that also includes transport fuels, heat and traditional biomass.

Evidence: The Electricity Act, 2003 governs the electricity chain, while the Ministry of Power Annual Report 2025-26 reports capacity, generation and supply outcomes separately.

UPSC trap: Do not equate a sanctioned plant, pipeline or connection with usable service.

Mains use: Use this boundary as the opening paragraph of an energy-infrastructure answer.

Recap: Energy policy succeeds when reliable and affordable service reaches the user, not when a capacity headline is announced. Electricity is only one carrier within a wider system that also includes transport fuels, heat and traditional biomass.

Core Session 02: Primary, secondary, final and useful energy

```
COAL / CRUDE / SUN / WIND / WATER / BIOMASS
-> ELECTRICITY / PETROL / HYDROGEN -> END USE -> SERVICE
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Primary energy exists before conversion; secondary energy is a converted carrier; final energy reaches the consumer; useful energy is the service after end-use losses.

Answer-grabbing line: Every energy statistic needs a stated stage in the conversion chain.

Must-write keywords: primary energy; secondary carrier; final energy; useful energy; conversion loss

Claim -> named evidence -> analysis -> qualification

- **Claim:** Every energy statistic needs a stated stage in the conversion chain.
- **Named evidence:** PPAC separates crude production, refinery output and petroleum-product consumption in its FY 2025-26 Ready Reckoner.
- **Analysis:** The stages prevent double counting and reveal whether loss or dependence arises at extraction, conversion, transport or end use.
- **Qualification:** Hydrogen is a carrier rather than a primary source; electricity can be produced from several primary sources.

Evidence: PPAC separates crude production, refinery output and petroleum-product consumption in its FY 2025-26 Ready Reckoner.

UPSC trap: Do not add primary fuel and the electricity generated from it into one undifferentiated total.

Mains use: Define the accounting boundary before comparing fuels.

Recap: Every energy statistic needs a stated stage in the conversion chain. Hydrogen is a carrier rather than a primary source; electricity can be produced from several primary sources.

Core Session 03: Commercial, non-commercial and modern energy

```
MARKETED: coal | petroleum | gas | grid power
NON-MARKETED: collected fuelwood | dung | residues
MODERN ACCESS: clean + controllable + reliable service
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Commercial energy is traded; non-commercial energy is often collected outside formal markets. Modern energy describes service quality and technology, not merely a cash transaction.

Answer-grabbing line: A connected household can remain energy-poor when supply, refill affordability or clean cooking is inadequate.

Must-write keywords: commercial energy; traditional biomass; clean cooking; time poverty; fuel stacking

Claim -> named evidence -> analysis -> qualification

- **Claim:** A connected household can remain energy-poor when supply, refill affordability or clean cooking is inadequate.
- **Named evidence:** SDG 7 and India's energy-access programmes distinguish connection from affordable, reliable and modern service.
- **Analysis:** Traditional fuels impose hidden health and time costs, disproportionately on women, while modern access supports education and enterprise.
- **Qualification:** Biomass can be modern and commercial when processed as pellets or biogas; categories are not identical.

Evidence: SDG 7 and India's energy-access programmes distinguish connection from affordable, reliable and modern service.

UPSC trap: Do not classify every renewable source as non-commercial.

Mains use: Link energy access to gender, health, education and productivity.

Recap: A connected household can remain energy-poor when supply, refill affordability or clean cooking is inadequate. Biomass can be modern and commercial when processed as pellets or biogas; categories are not identical.

Core Session 04: Energy mix and denominator discipline

```
ENERGY MIX MAY MEAN:
primary supply | installed power capacity | generation | final consumption
SAME LABEL, DIFFERENT DENOMINATOR
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: An energy mix is the share of sources within a defined aggregate; its denominator, unit, geography and period determine its meaning.

Answer-grabbing line: Capacity mix, generation mix and total-energy mix answer different questions.

Must-write keywords: energy mix; denominator; stock; flow; capacity share; generation share

Claim -> named evidence -> analysis -> qualification

- **Claim:** Capacity mix, generation mix and total-energy mix answer different questions.
- **Named evidence:** At 31 December 2025 non-fossil sources were 51.93% of installed power capacity; during April-December 2025 they supplied 30.41% of electricity generation, Ministry of Power Annual Report 2025-26.
- **Analysis:** Utilisation, weather, outages and dispatch explain why nameplate shares do not equal output shares.
- **Qualification:** The cited generation figure concerns electricity, not transport fuels or total primary energy.

Evidence: At 31 December 2025 non-fossil sources were 51.93% of installed power capacity; during April-December 2025 they supplied 30.41% of electricity generation, Ministry of Power Annual Report 2025-26.

UPSC trap: Never write that half of India's total energy was non-fossil from a capacity statistic.

Mains use: Attach unit, denominator, period and source to every mix claim.

Recap: Capacity mix, generation mix and total-energy mix answer different questions. The cited generation figure concerns electricity, not transport fuels or total primary energy.

Core Session 05: Energy security: six dimensions

```
AVAILABILITY -> ACCESS -> AFFORDABILITY
-> RELIABILITY -> SUSTAINABILITY -> RESILIENCE
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Energy security is sustained access to adequate and affordable energy with reliable infrastructure, environmental durability and recovery from physical, price, geopolitical, cyber and climate shocks.

Answer-grabbing line: Energy independence is not autarky; diversified interdependence can be safer than costly self-sufficiency.

Must-write keywords: availability; accessibility; affordability; reliability; sustainability; resilience

Claim -> named evidence -> analysis -> qualification

- **Claim:** Energy independence is not autarky; diversified interdependence can be safer than costly self-sufficiency.
- **Named evidence:** PPAC's FY 2025-26 provisional data report crude-oil import dependence of 88.7% and natural-gas import dependence of 50.1%, based on consumption.
- **Analysis:** Dependence transmits global price, exchange-rate and route shocks, but the percentage alone omits supplier diversity, stocks and substitution.
- **Qualification:** Each dependence ratio is commodity-specific and does not measure refinery strength or product exports.

Evidence: PPAC's FY 2025-26 provisional data report crude-oil import dependence of 88.7% and natural-gas import dependence of 50.1%, based on consumption.

UPSC trap: Do not use one import ratio as a complete security index.

Mains use: Evaluate every policy against all six dimensions.

Recap: Energy independence is not autarky; diversified interdependence can be safer than costly self-sufficiency. Each dependence ratio is commodity-specific and does not measure refinery strength or product exports.

Core Session 06: Electricity value chain

```
GENERATION -> CTU/STU TRANSMISSION -> SYSTEM OPERATION
-> DISTRIBUTION / OPEN ACCESS -> METER -> CONSUMER
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: The electricity chain separates generation, transmission, system operation, distribution, retail supply and consumption, even when ownership spans layers.

Answer-grabbing line: A failure at the financially weakest layer can interrupt the whole chain.

Must-write keywords: generation; transmission; system operation; distribution; retail; consumer

Claim -> named evidence -> analysis -> qualification

- **Claim:** A failure at the financially weakest layer can interrupt the whole chain.
- **Named evidence:** The Electricity Act, 2003 assigns distinct licensing, regulatory and operational roles across the chain.
- **Analysis:** Electricity must be scheduled through networks and retail revenue must return through the chain to generators.
- **Qualification:** Transmission ownership is not the same as neutral system operation; a DISCOM is not a generator merely because it buys power.

Evidence: The Electricity Act, 2003 assigns distinct licensing, regulatory and operational roles across the chain.

UPSC trap: Do not collapse Grid Controller, CTU and DISCOM into one institution.

Mains use: Trace electricity, money and accountability separately.

Recap: A failure at the financially weakest layer can interrupt the whole chain. Transmission ownership is not the same as neutral system operation; a DISCOM is not a generator merely because it buys power.

Core Session 07: MW, MWh and billion units

```
POWER: MW / GW = rate at an instant
ENERGY: MWh / GWh / BU = quantity over time
STORAGE: needs both MW and MWh
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Power measures a rate; energy measures the quantity delivered over time.

Answer-grabbing line: A 1,000 MW plant is a capability; its annual MWh depends on utilisation and availability.

Must-write keywords: MW; GW; MWh; GWh; BU; duration

Claim -> named evidence -> analysis -> qualification

- **Claim:** A 1,000 MW plant is a capability; its annual MWh depends on utilisation and availability.
- **Named evidence:** The Ministry of Power reports 513,729.69 MW installed at 31 December 2025 and 1,829.70 BU generated in FY 2024-25 as separate measures.
- **Analysis:** The same distinction determines whether storage can supply a brief frequency response or a multi-hour peak.
- **Qualification:** A 100 MW/400 MWh battery has about four hours of rated duration before efficiency and operating limits.

Evidence: The Ministry of Power reports 513,729.69 MW installed at 31 December 2025 and 1,829.70 BU generated in FY 2024-25 as separate measures.

UPSC trap: Never compare GW directly with BU.

Mains use: Convert the question into rate, quantity and time.

Recap: A 1,000 MW plant is a capability; its annual MWh depends on utilisation and availability. A 100 MW/400 MWh battery has about four hours of rated duration before efficiency and operating limits.

Core Session 08: Capacity factor, availability and PLF

```
POSSIBLE ENERGY = MW x HOURS
CAPACITY FACTOR = ACTUAL / POSSIBLE
PLF = thermal utilisation; EFFICIENCY is separate
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Capacity factor compares actual output with full-rating output; PLF is commonly used for thermal stations; availability asks whether the unit could run.

Answer-grabbing line: Low PLF may reflect demand, fuel, maintenance or dispatch, not only technical inefficiency.

Must-write keywords: capacity factor; PLF; availability; heat rate; utilisation

Claim -> named evidence -> analysis -> qualification

- **Claim:** Low PLF may reflect demand, fuel, maintenance or dispatch, not only technical inefficiency.
- **Named evidence:** CEA and Ministry of Power reports publish installed capacity, generation and thermal PLF as different indicators.
- **Analysis:** Fixed cost persists when a plant runs less, which can raise per-unit recovery and stranded-cost concern.
- **Qualification:** High PLF can coexist with poor heat rate; low solar capacity factor does not prove defective equipment.

Evidence: CEA and Ministry of Power reports publish installed capacity, generation and thermal PLF as different indicators.

UPSC trap: Do not call PLF an efficiency ratio.

Mains use: Use PLF to analyse utilisation, then identify the cause.

Recap: Low PLF may reflect demand, fuel, maintenance or dispatch, not only technical inefficiency. High PLF can coexist with poor heat rate; low solar capacity factor does not prove defective equipment.

Core Session 09: Peak demand, energy requirement and deficits

```
PEAK DEFICIT = unmet MW at maximum interval
ENERGY DEFICIT = unmet energy over a period
BOTH differ from local outage quality
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Peak demand is the highest instantaneous requirement; energy requirement is cumulative demand over time; their deficits measure different shortages.

Answer-grabbing line: Annual energy adequacy does not guarantee adequacy at the evening peak.

Must-write keywords: peak demand; peak met; energy requirement; energy supplied; unserved energy

Claim -> named evidence -> analysis -> qualification

- **Claim:** Annual energy adequacy does not guarantee adequacy at the evening peak.
- **Named evidence:** FY 2024-25 energy and peak deficits were 0.1% and 0.0%; CEA's provisional August 2026 report records all-India peak demand and peak met at 258,270 MW.
- **Analysis:** Very small national deficits can coexist with local outages, voltage problems, suppressed demand or distribution constraints.
- **Qualification:** Reported demand met does not measure unconstrained latent demand or every consumer's supply hours.

Evidence: FY 2024-25 energy and peak deficits were 0.1% and 0.0%; CEA's provisional August 2026 report records all-India peak demand and peak met at 258,270 MW.

UPSC trap: Do not infer local reliability from an all-India peak figure.

Mains use: State both peak and energy measures, then add quality evidence.

Recap: Annual energy adequacy does not guarantee adequacy at the evening peak. Reported demand met does not measure unconstrained latent demand or every consumer's supply hours.

Core Session 10: Load curve: base, intermediate and peak

```
LOAD DURATION CURVE
BASE: many hours | INTERMEDIATE: cycling | PEAK: scarce hours
TECHNOLOGY VALUE depends on duration + ramp + availability
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: A load curve orders demand across time; base, intermediate and peaking needs differ in duration, ramping and economic value.

Answer-grabbing line: The cheapest annual-average generator is not necessarily least-cost for every hour.

Must-write keywords: load curve; base load; intermediate load; peaking; ramp rate

Claim -> named evidence -> analysis -> qualification

- **Claim:** The cheapest annual-average generator is not necessarily least-cost for every hour.
- **Named evidence:** Hydro, storage, gas and demand response can provide flexibility even when their energy costs differ.
- **Analysis:** System planning chooses a portfolio that minimises total reliable-service cost across hours.
- **Qualification:** Base load is a demand-pattern concept, not a permanent legal label attached to one technology.

Evidence: Hydro, storage, gas and demand response can provide flexibility even when their energy costs differ.

UPSC trap: Do not say renewables replace baseload without discussing firmness and storage.

Mains use: Draw the load-duration logic before recommending a mix.

Recap: The cheapest annual-average generator is not necessarily least-cost for every hour. Base load is a demand-pattern concept, not a permanent legal label attached to one technology.

Core Session 11: Merit order and security-constrained dispatch

```
ELIGIBLE UNITS -> variable-cost order
+ congestion + reserve + ramp + minimum load -> DISPATCH
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Economic dispatch minimises short-run operating cost subject to grid security, contracts, ramping, minimum-load and congestion constraints.

Answer-grabbing line: Merit order is a rule inside a constrained system, not permission to ignore reliability.

Must-write keywords: variable cost; merit order; security constraint; ramping; technical minimum

Claim -> named evidence -> analysis -> qualification

- **Claim:** Merit order is a rule inside a constrained system, not permission to ignore reliability.
- **Named evidence:** CERC's grid and market framework combines scheduling, despatch, imbalance handling and congestion management.
- **Analysis:** Low-marginal-cost renewable output can displace thermal energy while flexible capacity remains valuable during scarcity.
- **Qualification:** PPA fixed charges and technical limits mean short-run dispatch does not settle total system cost.

Evidence: CERC's grid and market framework combines scheduling, despatch, imbalance handling and congestion management.

UPSC trap: Do not rank technologies only by fuel cost or LCOE.

Mains use: Compare marginal dispatch cost with adequacy and flexibility.

Recap: Merit order is a rule inside a constrained system, not permission to ignore reliability. PPA fixed charges and technical limits mean short-run dispatch does not settle total system cost.

Core Session 12: Real-time balancing and reserves

```
FORECAST -> DAY-AHEAD SCHEDULE -> REAL-TIME UPDATE
-> DEVIATION SETTLEMENT + RESERVES -> FREQUENCY CONTROL
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Power systems continuously match injections and withdrawals; reserves and balancing services correct forecast errors and contingencies.

Answer-grabbing line: Reliability is a valued operating service, not a free by-product of installed capacity.

Must-write keywords: frequency; reserve; balancing; schedule; contingency; black start

Claim -> named evidence -> analysis -> qualification

- **Claim:** Reliability is a valued operating service, not a free by-product of installed capacity.
- **Named evidence:** The Indian Electricity Grid Code, 2023 took effect on 1 October 2023; ancillary-service and DSM regulations complement it.
- **Analysis:** Forecast errors and outages impose balancing costs that should be signalled without replacing normal contracting.
- **Qualification:** Deviation settlement is a settlement and discipline mechanism, not an energy exchange.

Evidence: The Indian Electricity Grid Code, 2023 took effect on 1 October 2023; ancillary-service and DSM regulations complement it.

UPSC trap: Do not describe every unscheduled flow as an ancillary service.

Mains use: Explain schedule, deviation, corrective response and settlement.

Recap: Reliability is a valued operating service, not a free by-product of installed capacity. Deviation settlement is a settlement and discipline mechanism, not an energy exchange.

Core Session 13: Electricity Act, 2003 architecture

```
ACT 2003
GENERATION generally delicensed
TRANSMISSION / DISTRIBUTION / TRADING licensed
OPEN ACCESS + REGULATORS + APTEL + CONSUMER DUTIES
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: The Electricity Act, 2003 consolidated the law and created a competition-plus-regulation architecture.

Answer-grabbing line: The Act liberalised selected layers while retaining regulated networks and public-service obligations.

Must-write keywords: Electricity Act 2003; licensing; open access; tariff; APTEL; section 65

Claim -> named evidence -> analysis -> qualification

- **Claim:** The Act liberalised selected layers while retaining regulated networks and public-service obligations.
- **Named evidence:** The Act provides for CERC, SERCs, APTEL, national policies, grid discipline, tariff principles and transparent state subsidy.
- **Analysis:** Competition is easier in generation and trading than in monopoly wires, so regulation remains central.
- **Qualification:** Generation delicensing does not erase technical, hydro, environmental or grid approvals.

Evidence: The Act provides for CERC, SERCs, APTEL, national policies, grid discipline, tariff principles and transparent state subsidy.

UPSC trap: Do not describe the Act as complete privatisation.

Mains use: Organise legal answers by layer, regulator, competition and safeguard.

Recap: The Act liberalised selected layers while retaining regulated networks and public-service obligations. Generation delicensing does not erase technical, hydro, environmental or grid approvals.

Core Session 14: CEA: technical authority and planner

```
CEA -> data + technical standards + planning advice
NOT retail tariff regulator | NOT real-time system operator
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: The Central Electricity Authority is the statutory technical and planning body under the Electricity Act.

Answer-grabbing line: CEA supplies the engineering and evidence base for policy, planning and reliability.

Must-write keywords: CEA; standards; National Electricity Plan; statistics; planning

Claim -> named evidence -> analysis -> qualification

- **Claim:** CEA supplies the engineering and evidence base for policy, planning and reliability.
- **Named evidence:** CEA publishes installed-capacity, generation, power-supply and planning documents.
- **Analysis:** Demand forecasts, resource adequacy and transmission studies prevent target-led underplanning.
- **Qualification:** CEA advises and sets technical standards; commissions decide tariffs and disputes within mandate.

Evidence: CEA publishes installed-capacity, generation, power-supply and planning documents.

UPSC trap: Do not assign retail tariff setting to CEA.

Mains use: Cite CEA for technical data and name the actual decision-maker.

Recap: CEA supplies the engineering and evidence base for policy, planning and reliability. CEA advises and sets technical standards; commissions decide tariffs and disputes within mandate.

Core Session 15: CERC, SERCs and APTEL

```
INTER-STATE / CENTRAL -> CERC
INTRA-STATE / RETAIL -> SERC
APPEAL FROM COMMISSIONS -> APTEL
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Electricity regulation is divided mainly by inter-state/central and intra-state jurisdictions, with appeals to APTEL.

Answer-grabbing line: Jurisdiction follows the transaction and statute, not physical location alone.

Must-write keywords: CERC; SERC; APTEL; tariff; licence; adjudication

Claim -> named evidence -> analysis -> qualification

- **Claim:** Jurisdiction follows the transaction and statute, not physical location alone.
- **Named evidence:** The Electricity Act, 2003 establishes the commissions and appellate route.
- **Analysis:** CERC shapes inter-state transmission, central tariffs and wholesale markets; SERCs regulate distribution and retail tariffs.

- **Qualification:** Constitutional and statutory judicial review remains; APTEL is not the original tariff setter in every matter.

Evidence: The Electricity Act, 2003 establishes the commissions and appellate route.

UPSC trap: Do not call CERC the regulator of every retail tariff.

Mains use: Name transaction, jurisdiction and appeal route.

Recap: Jurisdiction follows the transaction and statute, not physical location alone. Constitutional and statutory judicial review remains; APTEL is not the original tariff setter in every matter.

Core Session 16: Grid Controller and load despatch

```
GRID CONTROLLER INDIA: NLDC + RLDCs
SLDCs: state operation
CTU/STU: network planning/access, not identical functions
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: System operation coordinates schedules, congestion, reserves and security independently of commercial ownership.

Answer-grabbing line: Neutral operation is the trust infrastructure of an interconnected market.

Must-write keywords: Grid-India; NLDC; RLDC; SLDC; CTU; STU

Claim -> named evidence -> analysis -> qualification

- **Claim:** Neutral operation is the trust infrastructure of an interconnected market.
- **Named evidence:** Grid Controller of India Limited operates national and regional load-despatch functions; SLDCs operate state systems.
- **Analysis:** A wider synchronous grid shares diversity and reserves but increases coordination and cyber-resilience needs.
- **Qualification:** Transmission utilities and load-despatch centres have connected but distinct roles.

Evidence: Grid Controller of India Limited operates national and regional load-despatch functions; SLDCs operate state systems.

UPSC trap: Do not use PGCIL, CTU and Grid-India interchangeably.

Mains use: Map owner, planner, operator and regulator separately.

Recap: Neutral operation is the trust infrastructure of an interconnected market. Transmission utilities and load-despatch centres have connected but distinct roles.

Core Session 17: Current amendment and rule status

```
ENACTED: Electricity Act, 2003 + notified rules
DRAFT: Electricity (Amendment) Bill, 2025
GREEN OPEN ACCESS: >=100 kW, subject to rules
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: The legal-status ladder separates an Act, notified rule/regulation, draft Bill, consultation and administrative direction.

Answer-grabbing line: A proposal does not amend the law until enacted and commenced.

Must-write keywords: legal status; draft Bill; notified rule; commencement; consultation

Claim -> named evidence -> analysis -> qualification

- **Claim:** A proposal does not amend the law until enacted and commenced.
- **Named evidence:** The Ministry circulated the draft Electricity (Amendment) Bill, 2025 in October-November 2025; no enactment was located by 10 September 2026. Green Energy Open Access Rules, 2022, as amended through 2024, use a 100 kW threshold.
- **Analysis:** Status discipline prevents proposed distribution reforms from being written as present law.

- **Qualification:** State implementation and commission orders still matter after a central rule is notified.

Evidence: The Ministry circulated the draft Electricity (Amendment) Bill, 2025 in October-November 2025; no enactment was located by 10 September 2026. Green Energy Open Access Rules, 2022, as amended through 2024, use a 100 kW threshold.

UPSC trap: Do not say the 2025 draft Bill replaced the 2003 Act.

Mains use: Date every instrument and label enacted, notified, proposed or draft.

Recap: A proposal does not amend the law until enacted and commenced. State implementation and commission orders still matter after a central rule is notified.

Core Session 18: Tariff components and cost to serve

```
TARIFF = power purchase + network/loss + O&M/finance
+ approved return +/- subsidy/cross-subsidy/deferred recovery
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: A tariff recovers approved costs and allocates them across consumer classes, time and fixed or variable charges.

Answer-grabbing line: The visible per-kWh price may hide fiscal subsidy, cross-subsidy or deferred recovery.

Must-write keywords: cost to serve; fixed charge; energy charge; fuel adjustment; regulatory asset

Claim -> named evidence -> analysis -> qualification

- **Claim:** The visible per-kWh price may hide fiscal subsidy, cross-subsidy or deferred recovery.
- **Named evidence:** SERC tariff orders separate power-purchase, network, employee/O&M, depreciation, interest and return elements.
- **Analysis:** Cost-reflective design supports investment, but abrupt correction can harm lifeline consumers and small firms.
- **Qualification:** Average cost can conceal time-of-day and location; tariff is not identical to marginal cost.

Evidence: SERC tariff orders separate power-purchase, network, employee/O&M, depreciation, interest and return elements.

UPSC trap: Do not treat a low tariff as proof of a low-cost system.

Mains use: Decompose who pays now, later, through taxes and through reliability.

Recap: The visible per-kWh price may hide fiscal subsidy, cross-subsidy or deferred recovery. Average cost can conceal time-of-day and location; tariff is not identical to marginal cost.

Core Session 19: Explicit subsidy and cross-subsidy

```
STATE BUDGET -> explicit subsidy -> DISCOM
HIGHER CLASS TARIFF -> cross-subsidy -> supported class
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Explicit subsidy is budget funded; cross-subsidy charges one consumer class above cost to support another.

Answer-grabbing line: Transparent subsidy preserves policy choice; delayed subsidy converts social policy into utility debt.

Must-write keywords: section 65; subsidy; cross-subsidy; incidence; lifeline tariff

Claim -> named evidence -> analysis -> qualification

- **Claim:** Transparent subsidy preserves policy choice; delayed subsidy converts social policy into utility debt.
- **Named evidence:** Section 65 of the Electricity Act requires state subsidy payment in the prescribed manner; tariff policy seeks gradual cross-subsidy reduction.
- **Analysis:** Excess industrial cross-subsidy can encourage captive or open-access exit and shrink the high-paying base.

- **Qualification:** Targeted support may remain justified; equal tariffs are not automatically equitable.

Evidence: Section 65 of the Electricity Act requires state subsidy payment in the prescribed manner; tariff policy seeks gradual cross-subsidy reduction.

UPSC trap: Do not merge tax-funded subsidy with cross-subsidy.

Mains use: Show fiscal incidence, consumer incidence and DISCOM cash effect.

Recap: Transparent subsidy preserves policy choice; delayed subsidy converts social policy into utility debt. Targeted support may remain justified; equal tariffs are not automatically equitable.

Core Session 20: Open access and network charges

```
ELIGIBLE USER -> chooses supplier
PAYS energy + transmission/wheeling + losses
+ lawful cross-subsidy/standby/additional charges
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Open access permits eligible users to use networks non-discriminatorily, subject to technical feasibility and applicable charges.

Answer-grabbing line: Choice in supply cannot mean free use of monopoly wires.

Must-write keywords: open access; wheeling; surcharge; standby; network cost

Claim -> named evidence -> analysis -> qualification

- **Claim:** Choice in supply cannot mean free use of monopoly wires.
- **Named evidence:** The Electricity Act and Green Energy Open Access Rules provide the legal route; CERC/SERC regulations operationalise it.
- **Analysis:** Open access can discipline procurement cost but may strand contracted capacity and shift fixed cost.
- **Qualification:** Captive, green and ordinary open access can have different conditions.

Evidence: The Electricity Act and Green Energy Open Access Rules provide the legal route; CERC/SERC regulations operationalise it.

UPSC trap: Do not describe open access as privatisation of the grid.

Mains use: Balance competition with network-cost recovery and universal service.

Recap: Choice in supply cannot mean free use of monopoly wires. Captive, green and ordinary open access can have different conditions.

Core Session 21: PPAs and procurement

```
DEMAND FORECAST -> TENDER / PPA
-> tariff + tenure + availability + fuel + change-in-law
-> payment security -> dispatch + settlement
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: A Power Purchase Agreement allocates price, volume, performance and regulatory risks between buyer and generator.

Answer-grabbing line: Long contracts create bankability but can lock consumers into wrong demand, fuel or technology assumptions.

Must-write keywords: PPA; fixed charge; variable charge; take-or-pay; change in law; payment security

Claim -> named evidence -> analysis -> qualification

- **Claim:** Long contracts create bankability but can lock consumers into wrong demand, fuel or technology assumptions.
- **Named evidence:** Competitive bidding and standard guidelines are used for several generation and storage procurements.

- **Analysis:** Well-designed PPAs lower financing cost; weak forecasts, transmission delay or renegotiation risk raise system cost.
- **Qualification:** A discovered tariff is not the delivered tariff after network, balancing, curtailment and payment risk.

Evidence: Competitive bidding and standard guidelines are used for several generation and storage procurements.

UPSC trap: Do not equate the lowest auction bid with full social cost.

Mains use: Audit forecast, risk allocation, transmission and payment.

Recap: Long contracts create bankability but can lock consumers into wrong demand, fuel or technology assumptions. A discovered tariff is not the delivered tariff after network, balancing, curtailment and payment risk.

Core Session 22: Power exchanges and short-term markets

```
DAY-AHEAD -> REAL-TIME -> TERM-AHEAD / GREEN SEGMENTS
BIDS -> CLEARING -> SCHEDULE -> SETTLEMENT
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Power exchanges provide organised short-term trading under CERC regulation; bilateral and long-term procurement continue alongside them.

Answer-grabbing line: Price discovery works only within transmission, liquidity, concentration and settlement constraints.

Must-write keywords: power exchange; day-ahead; real-time; term-ahead; clearing

Claim -> named evidence -> analysis -> qualification

- **Claim:** Price discovery works only within transmission, liquidity, concentration and settlement constraints.
- **Named evidence:** CERC's 23 July 2025 direction initiated phased work on market coupling, consultation and shadow-pilot preparation.
- **Analysis:** Short-term prices reveal scarcity and flexibility but can be volatile and cover only part of procurement.
- **Qualification:** Market coupling was not treated as universally completed at the cutoff.

Evidence: CERC's 23 July 2025 direction initiated phased work on market coupling, consultation and shadow-pilot preparation.

UPSC trap: Do not infer the whole system's average cost from one exchange price.

Mains use: State product, interval, market share and regulatory status.

Recap: Price discovery works only within transmission, liquidity, concentration and settlement constraints. Market coupling was not treated as universally completed at the cutoff.

Core Session 23: Scheduling and deviation settlement

```
FORECAST + DECLARED CAPABILITY -> SCHEDULE
ACTUAL - SCHEDULE = DEVIATION -> DSM CHARGE
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Scheduling commits injections and withdrawals; deviation settlement prices departures to support discipline and balance.

Answer-grabbing line: Deviation settlement corrects imbalance incentives; it is not a substitute for buying energy.

Must-write keywords: schedule; deviation; DSM; reference charge; settlement

Claim -> named evidence -> analysis -> qualification

- **Claim:** Deviation settlement corrects imbalance incentives; it is not a substitute for buying energy.
- **Named evidence:** CERC's DSM Regulations, 2024 operate with IEGC 2023; a draft third amendment was under consultation in 2026.
- **Analysis:** Charges influence forecasting and behaviour, especially for variable renewables and storage.
- **Qualification:** Not every deviation is misconduct; rules recognise technology and system conditions.

Evidence: CERC's DSM Regulations, 2024 operate with IEGC 2023; a draft third amendment was under consultation in 2026.

UPSC trap: Do not call DSM a penalty-only regime or exchange product.

Mains use: Explain schedule, measured deviation, charge and correction.

Recap: Deviation settlement corrects imbalance incentives; it is not a substitute for buying energy. Not every deviation is misconduct; rules recognise technology and system conditions.

Core Session 24: Ancillary services and resource adequacy

```
ENERGY MARKET -> MWh
ANCILLARY SERVICES -> response/reserve
RESOURCE ADEQUACY -> dependable capability ahead
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Ancillary services procure response and reserves; resource adequacy plans dependable capability for demand and uncertainty.

Answer-grabbing line: The scarce product in a renewable-rich grid is often response at the right time, not annual energy alone.

Must-write keywords: reserve; SRAS; TRAS; adequacy; dependable capacity

Claim -> named evidence -> analysis -> qualification

- **Claim:** The scarce product in a renewable-rich grid is often response at the right time, not annual energy alone.
- **Named evidence:** The Ministry of Power Annual Report 2025-26 identifies scheduling, imbalance, congestion and ancillary services as market-design pillars.
- **Analysis:** Separate products can remunerate flexibility that energy-only revenue may not cover.
- **Qualification:** Adequacy is not guaranteed dispatch, and ancillary service is not backup diesel by definition.

Evidence: The Ministry of Power Annual Report 2025-26 identifies scheduling, imbalance, congestion and ancillary services as market-design pillars.

UPSC trap: Do not add nameplate capacities without peak correlation and outages.

Mains use: Use dependable capacity and unserved-energy logic.

Recap: The scarce product in a renewable-rich grid is often response at the right time, not annual energy alone. Adequacy is not guaranteed dispatch, and ancillary service is not backup diesel by definition.

Core Session 25: DISCOM cash-flow model

```
INFLOW: bills + subsidy received + government dues
OUTFLOW: power + network/O&M + interest + capex
TIMING GAP -> borrowing + generator dues
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: DISCOM economics combines losses, tariff adequacy, collection, subsidy timing, procurement obligations and service quality.

Answer-grabbing line: Distribution is the commercial hinge: an insolvent last mile cannot turn generation into reliable service.

Must-write keywords: DISCOM; billing efficiency; collection; subsidy receivable; power dues

Claim -> named evidence -> analysis -> qualification

- **Claim:** Distribution is the commercial hinge: an insolvent last mile cannot turn generation into reliable service.
- **Named evidence:** RDSS and annual power reporting track operational and financial indicators.
- **Analysis:** Cash shortage reduces maintenance and procurement, worsening outages and collections.
- **Qualification:** Accounting profit can differ from cash flow; subsidy booked but unpaid cannot settle generator bills.

Evidence: RDSS and annual power reporting track operational and financial indicators.

UPSC trap: Do not diagnose every DISCOM loss as theft.

Mains use: Build diagnosis from energy account, tariff, subsidy and cash timing.

Recap: Distribution is the commercial hinge: an insolvent last mile cannot turn generation into reliable service. Accounting profit can differ from cash flow; subsidy booked but unpaid cannot settle generator bills.

Core Session 26: AT&C loss

```
INPUT ENERGY -> BILLED ENERGY -> COLLECTED REVENUE
AT&C = technical + commercial + collection failure
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: AT&C loss combines network loss with failure to meter, bill or collect the value of supplied energy.

Answer-grabbing line: Loss reduction requires feeder/transformer metering, network investment and commercial governance together.

Must-write keywords: AT&C; billing efficiency; collection efficiency; technical loss; commercial loss

Claim -> named evidence -> analysis -> qualification

- **Claim:** Loss reduction requires feeder/transformer metering, network investment and commercial governance together.
- **Named evidence:** Ministry of Power Annual Report 2025-26 reports national AT&C loss falling from 21.91% in FY 2020-21 to 15.04% in FY 2024-25.
- **Analysis:** Lower loss improves cash recovery and reduces purchased energy per billed unit.
- **Qualification:** State variation and accounting quality remain; a national percentage is not every DISCOM's result.

Evidence: Ministry of Power Annual Report 2025-26 reports national AT&C loss falling from 21.91% in FY 2020-21 to 15.04% in FY 2024-25.

UPSC trap: Do not define AT&C as theft alone.

Mains use: Pair AT&C with service and financial indicators.

Recap: Loss reduction requires feeder/transformer metering, network investment and commercial governance together. State variation and accounting quality remain; a national percentage is not every DISCOM's result.

Core Session 27: ACS-ARR, regulatory assets and payment discipline

```
ACS - ARR > 0 -> under-recovery per kWh
+ delayed subsidy/dues -> cash stress
+ regulatory asset -> future tariff + carrying cost
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: The ACS-ARR gap compares average supply cost with average realised revenue; regulatory assets defer approved recovery.

Answer-grabbing line: A near-zero booked gap is durable only if tariffs, subsidies, collections and power costs are timely and real.

Must-write keywords: ACS; ARR; regulatory asset; carrying cost; payment security

Claim -> named evidence -> analysis -> qualification

- **Claim:** A near-zero booked gap is durable only if tariffs, subsidies, collections and power costs are timely and real.
- **Named evidence:** The Ministry of Power Annual Report 2025-26 reports Rs 0.06/kWh for FY 2024-25, down from Rs 0.69/kWh in FY 2020-21.
- **Analysis:** Deferred cost and late subsidy can move stress across years without eliminating it.
- **Qualification:** Compare like definitions and examine subsidy cash received, not only accrued.

Evidence: The Ministry of Power Annual Report 2025-26 reports Rs 0.06/kWh for FY 2024-25, down from Rs 0.69/kWh in FY 2020-21.

UPSC trap: Do not infer financial health from AT&C alone.

Mains use: Use AT&C, ACS-ARR, payment timing and reliability together.

Recap: A near-zero booked gap is durable only if tariffs, subsidies, collections and power costs are timely and real. Compare like definitions and examine subsidy cash received, not only accrued.

Core Session 28: UDAY: debt restructuring and incentives

```
DISCOM DEBT at 30 Sep 2015 -> states take 75%
-> lower interest burden + operational commitments
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: UDAY, launched 20 November 2015, combined state debt takeover with operational and cost-reduction commitments.

Answer-grabbing line: Debt relief creates breathing space; it does not replace tariff, loss, procurement and governance reform.

Must-write keywords: UDAY; state debt takeover; interest burden; operational milestone

Claim -> named evidence -> analysis -> qualification

- **Claim:** Debt relief creates breathing space; it does not replace tariff, loss, procurement and governance reform.
- **Named evidence:** Participating states were to take over 75% of specified DISCOM debt outstanding at 30 September 2015.
- **Analysis:** Lower interest cost can help, but recurring under-recovery rebuilds liabilities.
- **Qualification:** UDAY was a policy arrangement, not the same scheme as RDSS.

Evidence: Participating states were to take over 75% of specified DISCOM debt outstanding at 30 September 2015.

UPSC trap: Do not claim debt transfer automatically reduced losses.

Mains use: Evaluate balance-sheet reset and post-reset incentives.

Recap: Debt relief creates breathing space; it does not replace tariff, loss, procurement and governance reform. UDAY was a policy arrangement, not the same scheme as RDSS.

Core Session 29: RDSS: conditional distribution reform

```
BASELINE + ACTION PLAN -> network works / smart meters
-> performance conditions -> funding -> service outcome
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: RDSS links infrastructure and smart-meter support with performance and reform conditions.

Answer-grabbing line: Technology enables accountability only when data, billing, tariffs and service systems work.

Must-write keywords: RDSS; smart prepaid meter; conditional funding; feeder segregation

Claim -> named evidence -> analysis -> qualification

- **Claim:** Technology enables accountability only when data, billing, tariffs and service systems work.
- **Named evidence:** The annual report states targets of 12-15% AT&C loss and zero ACS-ARR gap, sunset 31 March 2028, and 5.28 crore smart meters installed under various schemes by December 2025.
- **Analysis:** Meters can reduce reading and collection frictions and enable time signals.
- **Qualification:** Installed meter count is not proof of working communications, correct bills or grievance resolution.

Evidence: The annual report states targets of 12-15% AT&C loss and zero ACS-ARR gap, sunset 31 March 2028, and 5.28 crore smart meters installed under various schemes by December 2025.

UPSC trap: Do not call RDSS a mere renaming of UDAY.

Mains use: Judge RDSS by cash recovery, outages and consumer outcomes.

Recap: Technology enables accountability only when data, billing, tariffs and service systems work. Installed meter count is not proof of working communications, correct bills or grievance resolution.

Core Session 30: Coal value chain and quality

```
MINE -> WASH/GRADE -> RAIL/ROAD/PORT
-> PLANT STOCK -> COMBUSTION -> ASH -> CLOSURE
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Coal economics includes extraction, grade, transport, inventory, combustion, pollution control, ash use and closure.

Answer-grabbing line: Pithead production is not useful supply until the right grade reaches the right user.

Must-write keywords: thermal coal; coking coal; GCV; evacuation; pithead stock

Claim -> named evidence -> analysis -> qualification

- **Claim:** Pithead production is not useful supply until the right grade reaches the right user.
- **Named evidence:** The Ministry of Coal reports 1,047.523 MT all-India production in FY 2024-25.
- **Analysis:** Transport and quality affect heat rate, generation cost and plant availability.
- **Qualification:** A geological resource is not an economically recoverable reserve or dispatched coal.

Evidence: The Ministry of Coal reports 1,047.523 MT all-India production in FY 2024-25.

UPSC trap: Do not equate production with power-sector availability.

Mains use: Trace quantity, quality, logistics and liabilities.

Recap: Pithead production is not useful supply until the right grade reaches the right user. A geological resource is not an economically recoverable reserve or dispatched coal.

Core Session 31: Coal allocation, linkages and auctions

```
MINE AUCTION -> right to develop
LINKAGE / SHAKTI -> supply arrangement
E-AUCTION -> sale channel
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Coal-block auctions, commercial mining, linkages and e-auctions solve different access problems.

Answer-grabbing line: Transparent allocation does not remove geology, evacuation, clearance or demand risk.

Must-write keywords: commercial mining; linkage; SHAKTI; auction; FSA

Claim -> named evidence -> analysis -> qualification

- **Claim:** Transparent allocation does not remove geology, evacuation, clearance or demand risk.
- **Named evidence:** The Union Cabinet approved revised SHAKTI policy on 7 May 2025 for power-sector coal allocation.
- **Analysis:** Linkages support fuel security and finance; auctions allocate scarce rights or supplies through bidding.
- **Qualification:** An auction winner is not an operational mine or delivered fuel.

Evidence: The Union Cabinet approved revised SHAKTI policy on 7 May 2025 for power-sector coal allocation.

UPSC trap: Do not confuse a mine auction with a fuel-supply linkage.

Mains use: State right, contract, production, dispatch and use separately.

Recap: Transparent allocation does not remove geology, evacuation, clearance or demand risk. An auction winner is not an operational mine or delivered fuel.

Core Session 32: Coal imports and logistics

```
QUALITY / QUANTITY / COASTAL COST GAP
-> IMPORT -> PORT -> RAIL/COASTAL MOVEMENT -> USER
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Coal imports may reflect shortage, coking quality, coastal logistics or commercial choice.

Answer-grabbing line: Import substitution is efficient only when suitable domestic coal can be delivered at lower system cost.

Must-write keywords: coking coal; non-coking; import substitution; logistics; blending

Claim -> named evidence -> analysis -> qualification

- **Claim:** Import substitution is efficient only when suitable domestic coal can be delivered at lower system cost.
- **Named evidence:** Ministry of Coal reports FY 2025-26 imports of 246.37 MT: 66.33 MT coking and 180.04 MT non-coking.
- **Analysis:** Coking dependence differs from thermal dependence; delivered cost matters for coastal plants.
- **Qualification:** Import flows do not prove reserve scarcity or uniform avoidability.

Evidence: Ministry of Coal reports FY 2025-26 imports of 246.37 MT: 66.33 MT coking and 180.04 MT non-coking.

UPSC trap: Do not call every imported tonne avoidable.

Mains use: Separate quality, location, quantity and price drivers.

Recap: Import substitution is efficient only when suitable domestic coal can be delivered at lower system cost. Import flows do not prove reserve scarcity or uniform avoidability.

Core Session 33: Thermal power economics and flexibility

```
ANNUAL COST = fixed capacity + fuel/transport + O&M
+ environmental compliance + finance
VALUE = energy + dependable capacity + ramping
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Coal plants combine sunk capital with fuel, logistics and environmental costs, while also supplying dispatchable capacity.

Answer-grabbing line: Thermal value increasingly includes flexibility, not only annual energy.

Must-write keywords: fixed charge; variable charge; heat rate; ramping; technical minimum

Claim -> named evidence -> analysis -> qualification

- **Claim:** Thermal value increasingly includes flexibility, not only annual energy.
- **Named evidence:** At 31 December 2025 coal capacity was 219,610 MW; coal supplied 66.46% of April-December 2025 generation.
- **Analysis:** Lower running hours spread fixed cost over fewer units and increase flexibility requirements.
- **Qualification:** Coal generation share is not coal's share in total primary energy; unit performance varies.

Evidence: At 31 December 2025 coal capacity was 219,610 MW; coal supplied 66.46% of April-December 2025 generation.

UPSC trap: Do not assume all thermal plants are equally efficient or flexible.

Mains use: Compare reliability value with fuel, pollution and stranded risk.

Recap: Thermal value increasingly includes flexibility, not only annual energy. Coal generation share is not coal's share in total primary energy; unit performance varies.

Core Session 34: Thermal environmental cost and stranded assets

```
AIR + WATER + ASH + CARBON + LAND
versus FIRMNESS + RAMP
POLICY/TECH CHANGE -> lower use -> unrecovered debt
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Thermal appraisal must value local pollution, water, ash, mining and climate costs and recognise assets that may lose value early.

Answer-grabbing line: A credible transition neither ignores coal externalities nor retires firm service before substitutes exist.

Must-write keywords: FGD; water stress; ash; externality; stranded asset; closure

Claim -> named evidence -> analysis -> qualification

- **Claim:** A credible transition neither ignores coal externalities nor retires firm service before substitutes exist.
- **Named evidence:** Long-lived plants, mines, railways and PPAs create path dependence; the 2023 objective route tests location-specific water choices at coal stations.
- **Analysis:** Retrofit and flexible operation can preserve service but cost money; early closure can crystallise debt and regional loss.
- **Qualification:** Low PLF alone does not prove stranding because seasonal reserve value may remain.

Evidence: Long-lived plants, mines, railways and PPAs create path dependence; the 2023 objective route tests location-specific water choices at coal stations.

UPSC trap: Do not use 'baseload' to exempt environmental appraisal.

Mains use: Put reliability, external cost and transition liability together.

Recap: A credible transition neither ignores coal externalities nor retires firm service before substitutes exist. Low PLF alone does not prove stranding because seasonal reserve value may remain.

Core Session 35: Oil value chain and OMCs

```
UPSTREAM exploration/production -> MIDSTREAM shipping/storage
-> DOWNSTREAM refining/marketing/retail -> CONSUMER PRICE
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: The petroleum chain separates production from transport/storage and refining/marketing; OMCs operate mainly in downstream refining and markets.

Answer-grabbing line: Crude vulnerability depends on sourcing, routes, refining configuration, currency and product demand together.

Must-write keywords: upstream; midstream; downstream; refinery; product slate; OMC

Claim -> named evidence -> analysis -> qualification

- **Claim:** Crude vulnerability depends on sourcing, routes, refining configuration, currency and product demand together.
- **Named evidence:** PPAC FY 2025-26 reports domestic crude production of 28.0 MMT and product consumption of 241.6 MMT.
- **Analysis:** Complex refineries can process varied crude and export products even with high crude dependence.
- **Qualification:** Product consumption and crude production are different categories and cannot be directly subtracted as a balance.

Evidence: PPAC FY 2025-26 reports domestic crude production of 28.0 MMT and product consumption of 241.6 MMT.

UPSC trap: Do not call product exports evidence of crude self-sufficiency.

Mains use: Trace the molecule, foreign exchange and price through the chain.

Recap: Crude vulnerability depends on sourcing, routes, refining configuration, currency and product demand together. Product consumption and crude production are different categories and cannot be directly subtracted as a balance.

Core Session 36: Petroleum pricing and marketing

```
GLOBAL PRICE + EXCHANGE RATE + REFINING/MARKETING/FREIGHT
+ CENTRAL/STATE TAX +/- TARGETED SUPPORT -> RETAIL PRICE
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Retail fuel prices combine international conditions, exchange rates, costs, taxes and policy interventions.

Answer-grabbing line: A pump price is a fiscal and market composite, not a direct copy of crude price.

Must-write keywords: dynamic pricing; excise; VAT; under-recovery; subsidy

Claim -> named evidence -> analysis -> qualification

- **Claim:** A pump price is a fiscal and market composite, not a direct copy of crude price.
- **Named evidence:** PPAC publishes retail price build-ups and subsidy/under-recovery data; petrol and diesel are officially market-determined, while LPG support is targeted under current policy.
- **Analysis:** Tax cuts cushion inflation but lose revenue; broad price suppression creates fiscal cost and weakens conservation.
- **Qualification:** Market-determined does not mean taxes or public-sector choices are absent.

Evidence: PPAC publishes retail price build-ups and subsidy/under-recovery data; petrol and diesel are officially market-determined, while LPG support is targeted under current policy.

UPSC trap: Do not attribute every retail movement to crude alone.

Mains use: Decompose world price, rupee, tax and marketing components.

Recap: A pump price is a fiscal and market composite, not a direct copy of crude price. Market-determined does not mean taxes or public-sector choices are absent.

Core Session 37: PNGRB and pipelines

```
PNGRB: refining/processing/storage/transport/distribution/marketing/sale
EXCLUSION: production of crude oil and natural gas
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: PNGRB is the statutory downstream and network regulator under the PNGRB Act, 2006.

Answer-grabbing line: The regulator's perimeter follows activity, with upstream production expressly excluded.

Must-write keywords: PNGRB; common carrier; contract carrier; CGD; authorisation

Claim -> named evidence -> analysis -> qualification

- **Claim:** The regulator's perimeter follows activity, with upstream production expressly excluded.
- **Named evidence:** PNGRB's official site states the regulated activities and exclusion of crude-oil and natural-gas production.
- **Analysis:** Access and tariff regulation address natural-monopoly and network-expansion problems.
- **Qualification:** PNGRB does not set every retail fuel price or allocate upstream exploration blocks.

Evidence: PNGRB's official site states the regulated activities and exclusion of crude-oil and natural-gas production.

UPSC trap: Do not confuse PNGRB with DGH or MoPNG.

Mains use: Use the objective-test method: activity first, statutory perimeter second.

Recap: The regulator's perimeter follows activity, with upstream production expressly excluded. PNGRB does not set every retail fuel price or allocate upstream exploration blocks.

Core Session 38: Natural gas, LNG and city gas

```
DOMESTIC GAS + LNG IMPORT -> REGAS TERMINAL -> TRUNK PIPE
-> CGD / FERTILISER / POWER / INDUSTRY
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Gas infrastructure requires supply, regasification, pipelines, anchor demand and a competitive delivered price.

Answer-grabbing line: A pipeline without throughput is stranded capital; demand without a pipeline may never emerge.

Must-write keywords: LNG; regasification; pipeline; CGD; CNG; PNG; unified tariff

Claim -> named evidence -> analysis -> qualification

- **Claim:** A pipeline without throughput is stranded capital; demand without a pipeline may never emerge.
- **Named evidence:** PPAC FY 2025-26 provisional data report net production 34,326 MMSCM, LNG imports 34,427 MMSCM and consumption 68,753 MMSCM.
- **Analysis:** Imported LNG adds flexibility but exposes users to global prices; unified tariffs seek wider network access.
- **Qualification:** Gas has lower combustion pollution than coal in several dimensions but is not zero-carbon; methane leakage matters.

Evidence: PPAC FY 2025-26 provisional data report net production 34,326 MMSCM, LNG imports 34,427 MMSCM and consumption 68,753 MMSCM.

UPSC trap: Do not equate CGD authorisation with completed connections or affordable gas.

Mains use: Evaluate source, terminal, pipe, tariff and anchor load.

Recap: A pipeline without throughput is stranded capital; demand without a pipeline may never emerge. Gas has lower combustion pollution than coal in several dimensions but is not zero-carbon; methane leakage matters.

Core Session 39: Oil and gas dependence and resilience

```
DEPENDENCE RATIO + supplier concentration + route risk
+ contracts + stocks + substitution -> SECURITY
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Import dependence measures the consumption share met by imports; resilience also depends on concentration, route, contracts, stocks and alternatives.

Answer-grabbing line: A high dependence ratio signals exposure but not disruption probability or damage.

Must-write keywords: import dependence; concentration; route; term contract; spot cargo

Claim -> named evidence -> analysis -> qualification

- **Claim:** A high dependence ratio signals exposure but not disruption probability or damage.
- **Named evidence:** PPAC FY 2025-26 provisional dependence was 88.7% for crude and 50.1% for gas.
- **Analysis:** Long-term contracts reduce volume uncertainty; spot cargoes add flexibility; efficiency and electrification reduce demand.
- **Qualification:** Diversified suppliers may still share one chokepoint; each ratio uses a separate commodity formula.

Evidence: PPAC FY 2025-26 provisional dependence was 88.7% for crude and 50.1% for gas.

UPSC trap: Do not compare crude and gas percentages without denominators.

Mains use: Pair dependence with route, contract, stock and substitution.

Recap: A high dependence ratio signals exposure but not disruption probability or damage. Diversified suppliers may still share one chokepoint; each ratio uses a separate commodity formula.

Core Session 40: Strategic petroleum reserves and hedging

```
COMMERCIAL STOCK + STRATEGIC RESERVE + DIVERSE SOURCES/ROUTES
+ CONTRACT PORTFOLIO + FINANCIAL HEDGE
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Resilience combines physical inventories with diversified supply and price-risk management.

Answer-grabbing line: A reserve buys time; it does not create permanent domestic supply.

Must-write keywords: SPR; inventory; drawdown; diversification; hedge

Claim -> named evidence -> analysis -> qualification

- **Claim:** A reserve buys time; it does not create permanent domestic supply.
- **Named evidence:** Phase-I strategic reserve capacity is 5.33 MMT at Visakhapatnam, Mangaluru and Padur; a 6.5 MMT Phase-II at Chandikhol and Padur was approved in July 2021.
- **Analysis:** Stocks bridge disruption; diversification reduces recurrence; hedges can reduce price volatility.
- **Qualification:** Approved capacity is not operating or full inventory, and financial hedges cannot deliver physical crude.

Evidence: Phase-I strategic reserve capacity is 5.33 MMT at Visakhapatnam, Mangaluru and Padur; a 6.5 MMT Phase-II at Chandikhol and Padur was approved in July 2021.

UPSC trap: Do not state fixed days of cover without dated consumption and fill data.

Mains use: Use a layered resilience portfolio.

Recap: A reserve buys time; it does not create permanent domestic supply. Approved capacity is not operating or full inventory, and financial hedges cannot deliver physical crude.

Core Session 41: Renewable portfolio

```
SOLAR: daytime variable | WIND: seasonal/site variable
HYDRO: flexible/hydrology constrained | BIOMASS: feedstock constrained
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Renewable technologies have different temporal, spatial, network and ecological characteristics.

Answer-grabbing line: Renewable capacity is a portfolio, not a homogeneous block of megawatts.

Must-write keywords: solar; wind; hydro; biomass; variable renewable; dispatchable renewable

Claim -> named evidence -> analysis -> qualification

- **Claim:** Renewable capacity is a portfolio, not a homogeneous block of megawatts.
- **Named evidence:** At 31 December 2025 capacity was solar 135.81 GW, wind 54.51 GW, hydro 50.91 GW, and bio/waste about 11.61 GW.
- **Analysis:** Complementary profiles and geography can smooth output, while common weather can still correlate it.
- **Qualification:** Large hydro and nuclear are non-fossil; renewable and non-fossil are not identical.

Evidence: At 31 December 2025 capacity was solar 135.81 GW, wind 54.51 GW, hydro 50.91 GW, and bio/waste about 11.61 GW.

UPSC trap: Do not infer generation ranking from installed capacity.

Mains use: Compare profile, firmness, land, transmission and system cost.

Recap: Renewable capacity is a portfolio, not a homogeneous block of megawatts. Large hydro and nuclear are non-fossil; renewable and non-fossil are not identical.

Core Session 42: Renewable auctions and project status

```
RESOURCE + LAND + GRID + PPA -> REVERSE AUCTION
-> AWARD -> FINANCE -> COMMISSION -> GENERATION
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Renewable auctions discover a contracted tariff subject to tender design and risk allocation.

Answer-grabbing line: A low bid is credible only if land, equipment, finance, grid and buyer payment are deliverable.

Must-write keywords: reverse auction; bid tariff; solar park; SECI; payment security

Claim -> named evidence -> analysis -> qualification

- **Claim:** A low bid is credible only if land, equipment, finance, grid and buyer payment are deliverable.
- **Named evidence:** Central and state agencies use competitive bidding under notified frameworks.
- **Analysis:** Competition can lower tariff but aggressive bids can face delay, renegotiation or curtailment.
- **Qualification:** Auction tariff excludes several network, balancing and retail costs.

Evidence: Central and state agencies use competitive bidding under notified frameworks.

UPSC trap: Do not label tendered or awarded capacity as commissioned.

Mains use: Follow the complete project-status ladder.

Recap: A low bid is credible only if land, equipment, finance, grid and buyer payment are deliverable. Auction tariff excludes several network, balancing and retail costs.

Core Session 43: Intermittency, curtailment and grid integration

```
WEATHER -> VARIABLE OUTPUT -> forecast error / congestion / ramp
FORECAST + FLEXIBILITY + STORAGE + DEMAND RESPONSE -> RELIABLE SERVICE
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Variability is resource-driven change; curtailment is available generation deliberately not accepted.

Answer-grabbing line: Integration cost depends on when and where energy arrives, not only plant-level cost.

Must-write keywords: forecasting; curtailment; duck curve; ramp; balancing area

Claim -> named evidence -> analysis -> qualification

- **Claim:** Integration cost depends on when and where energy arrives, not only plant-level cost.
- **Named evidence:** The capacity-generation contrast at end-December and April-December 2025 illustrates why nameplate capacity is not output.
- **Analysis:** Transmission, storage, flexible supply, demand response and wider balancing areas reduce curtailment and scarcity ramps.
- **Qualification:** Curtailment may arise from congestion or security and is not automatically unlawful.

Evidence: The capacity-generation contrast at end-December and April-December 2025 illustrates why nameplate capacity is not output.

UPSC trap: Do not call every fall in solar output curtailment.

Mains use: Identify time, location, forecast, network and flexibility.

Recap: Integration cost depends on when and where energy arrives, not only plant-level cost. Curtailment may arise from congestion or security and is not automatically unlawful.

Core Session 44: Green Energy Corridors and planning

```
RE-RICH ZONE -> POOLING STATION -> HIGH-VOLTAGE CORRIDOR
-> STATE/INTERSTATE GRID -> DEMAND CENTRE
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Transmission must be planned ahead because lines often take longer than generation projects.

Answer-grabbing line: Generation without evacuation is stranded energy; wires without committed supply are stranded capital.

Must-write keywords: Green Energy Corridor; pooling station; CTU; STU; evacuation

Claim -> named evidence -> analysis -> qualification

- **Claim:** Generation without evacuation is stranded energy; wires without committed supply are stranded capital.
- **Named evidence:** MNRE GEC-II targets about 10,750 circuit-km and 27,500 MVA to integrate about 20 GW in seven states, at Rs 12,031.33 crore with 33% central assistance.
- **Analysis:** Corridors expand balancing and reduce congestion while creating route, land and cost-allocation questions.
- **Qualification:** Scheme design figures are not completed-line statistics.

Evidence: MNRE GEC-II targets about 10,750 circuit-km and 27,500 MVA to integrate about 20 GW in seven states, at Rs 12,031.33 crore with 33% central assistance.

UPSC trap: Do not treat a transmission sanction as generation.

Mains use: Link resource zone, line, market access and safeguards.

Recap: Generation without evacuation is stranded energy; wires without committed supply are stranded capital. Scheme design figures are not completed-line statistics.

Core Session 45: Battery and pumped-hydro storage

```
POWER MW x DURATION h = ENERGY MWh
CHARGE -> loss -> STORE -> DISCHARGE
BESS modular | PSP site/gestation constrained
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Storage shifts electricity through time and can provide reserves, ramping, congestion relief and backup.

Answer-grabbing line: Storage is not a source; it returns less energy than it absorbs while adding timing value.

Must-write keywords: round-trip efficiency; duration; state of charge; degradation; PSP; BESS

Claim -> named evidence -> analysis -> qualification

- **Claim:** Storage is not a source; it returns less energy than it absorbs while adding timing value.
- **Named evidence:** Economic Survey 2025-26 cites about 336 GWh needed by 2029-30 and 411 GWh by 2031-32; two VGF schemes launched in March 2024 and June 2025 support about 43 GWh.
- **Analysis:** Revenue stacking can improve viability across arbitrage, capacity and ancillary services.
- **Qualification:** Avoid double payment; batteries have mineral/recycling costs and PSP has land, water and ecology constraints.

Evidence: Economic Survey 2025-26 cites about 336 GWh needed by 2029-30 and 411 GWh by 2031-32; two VGF schemes launched in March 2024 and June 2025 support about 43 GWh.

UPSC trap: Do not compare storage MW with renewable GW without duration.

Mains use: State MW, MWh, duration, cycles, efficiency and service.

Recap: Storage is not a source; it returns less energy than it absorbs while adding timing value. Avoid double payment; batteries have mineral/recycling costs and PSP has land, water and ecology constraints.

Core Session 46: Green hydrogen economics

```
RE POWER + WATER -> ELECTROLYSER -> H2
-> FERTILISER / REFINERY / STEEL / SHIPPING
COST = power + electrolyser + finance + transport/storage
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Green hydrogen is hydrogen produced through electrolysis using renewable electricity under the applicable standard.

Answer-grabbing line: Use hydrogen first where direct electrification is difficult and industrial demand can anchor scale.

Must-write keywords: green hydrogen; electrolyser; SIGHT; additionality; hard-to-abate

Claim -> named evidence -> analysis -> qualification

- **Claim:** Use hydrogen first where direct electrification is difficult and industrial demand can anchor scale.
- **Named evidence:** The January 2023 mission has Rs 19,744 crore outlay and at least 5 MMT/year 2030 target; Economic Survey 2025-26 reports 862,000 tonnes/year allocated, 3,000 MW/year electrolyser capacity awarded and three port hubs.
- **Analysis:** Policy combines production support, manufacturing scale, demand creation and infrastructure.
- **Qualification:** Allocated or awarded capacity is not commissioned output; conversion loses energy.

Evidence: The January 2023 mission has Rs 19,744 crore outlay and at least 5 MMT/year 2030 target; Economic Survey 2025-26 reports 862,000 tonnes/year allocated, 3,000 MW/year electrolyser capacity awarded and three port hubs.

UPSC trap: Do not treat hydrogen as a primary source or universal substitute.

Mains use: Prioritise no-regret uses and separate target, award and output.

Recap: Use hydrogen first where direct electrification is difficult and industrial demand can anchor scale. Allocated or awarded capacity is not commissioned output; conversion loses energy.

Core Session 47: Nuclear power: bounded role

```
URANIUM -> FISSION HEAT -> STEAM -> ELECTRICITY
FIRM LOW-CARBON OUTPUT versus capital + safety + liability + waste
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Nuclear supplies firm low-carbon electricity but requires specialised safety, fuel-cycle, liability and waste institutions.

Answer-grabbing line: Its economic case is firmness and low operating carbon, not zero risk or zero cost.

Must-write keywords: nuclear; firm capacity; liability; waste; SMR

Claim -> named evidence -> analysis -> qualification

- **Claim:** Its economic case is firmness and low operating carbon, not zero risk or zero cost.
- **Named evidence:** Installed nuclear capacity was 8.78 GW at 31 December 2025; reactor technology detail belongs to Science and Technology.
- **Analysis:** Nuclear can complement variable renewables but faces high capital and long-construction risk.
- **Qualification:** Installed capacity does not reveal lifecycle cost or completed generation.

Evidence: Installed nuclear capacity was 8.78 GW at 31 December 2025; reactor technology detail belongs to Science and Technology.

UPSC trap: Do not make nuclear a substitute for every grid, storage or efficiency need.

Mains use: Keep treatment bounded to system value, finance and governance.

Recap: Its economic case is firmness and low operating carbon, not zero risk or zero cost. Installed capacity does not reveal lifecycle cost or completed generation.

Core Session 48: RPO, RCO, ESO, REC and carbon markets

```
RPO/RCO -> renewable-consumption obligation
REC -> 1 MWh renewable attribute | ESO -> storage share
PAT ESCert -> energy saving | CCTS credit -> GHG performance
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: These instruments have different legal bases, obligated entities, units and objectives.

Answer-grabbing line: A renewable attribute certificate is not automatically one tonne of carbon dioxide equivalent.

Must-write keywords: RPO; RCO; ESO; REC; ESCert; CCTS; double counting

Claim -> named evidence -> analysis -> qualification

- **Claim:** A renewable attribute certificate is not automatically one tonne of carbon dioxide equivalent.
- **Named evidence:** CERC REC Regulations, 2022 set one REC at one MWh; the ESO trajectory rises from 1% in FY 2023-24 to 4% in FY 2029-30; the 2022 Energy Conservation amendment enabled carbon trading and CCTS was notified in 2023.
- **Analysis:** Obligations create demand; certificates separate attributes from physical power or verified performance.
- **Qualification:** Eligibility, multipliers, banking and transition rules can change; instruments are not automatically fungible.

Evidence: CERC REC Regulations, 2022 set one REC at one MWh; the ESO trajectory rises from 1% in FY 2023-24 to 4% in FY 2029-30; the 2022 Energy Conservation amendment enabled carbon trading and CCTS was notified in 2023.

UPSC trap: Do not count one REC as one carbon credit.

Mains use: State statute, obligated entity, unit and compliance route.

Recap: A renewable attribute certificate is not automatically one tonne of carbon dioxide equivalent. Eligibility, multipliers, banking and transition rules can change; instruments are not automatically fungible.

Core Session 49: Energy efficiency: PAT, standards and labels

```
SERVICE DEMAND -> efficient process/appliance/building
-> lower energy per service -> bills/imports/emissions
subject to REBOUND
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Efficiency reduces energy input per service; conservation may reduce the amount of service used.

Answer-grabbing line: The cheapest unit is often the one not required, but savings need a credible baseline.

Must-write keywords: BEE; PAT; ESCert; star label; building code; rebound

Claim -> named evidence -> analysis -> qualification

- **Claim:** The cheapest unit is often the one not required, but savings need a credible baseline.
- **Named evidence:** The Energy Conservation Act underpins BEE's PAT, standards and labelling and building-efficiency programmes.
- **Analysis:** Efficiency can postpone generation and network investment while lowering bills and import demand.
- **Qualification:** Engineering savings may be offset by rebound, output growth or weak compliance; ESCerts are not RECs.

Evidence: The Energy Conservation Act underpins BEE's PAT, standards and labelling and building-efficiency programmes.

UPSC trap: Do not call conservation and efficiency identical.

Mains use: State baseline, normalisation, saving, cost and rebound.

Recap: The cheapest unit is often the one not required, but savings need a credible baseline. Engineering savings may be offset by rebound, output growth or weak compliance; ESCerts are not RECs.

Core Session 50: Demand-side management and time signals

```
FLAT TARIFF -> weak timing signal
TIME-OF-DAY + DEMAND RESPONSE -> shift load
-> absorb solar + reduce peak
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Demand-side management changes the level or timing of use through information, automation, pricing or contracted response.

Answer-grabbing line: Flexible demand can provide balancing value at lower cost than new supply.

Must-write keywords: time-of-day tariff; demand response; smart meter; peak shaving

Claim -> named evidence -> analysis -> qualification

- **Claim:** Flexible demand can provide balancing value at lower cost than new supply.
- **Named evidence:** RDSS smart metering and tariff frameworks enable granular consumption and time signals.
- **Analysis:** Shifting pumps, cooling, industry or EV charging can absorb midday solar and reduce evening ramps.
- **Qualification:** Poorly designed tariffs can burden users unable to shift; privacy and grievance safeguards matter.

Evidence: RDSS smart metering and tariff frameworks enable granular consumption and time signals.

UPSC trap: Do not confuse demand-side management with deviation settlement, also abbreviated DSM.

Mains use: Spell out the abbreviation and identify the mechanism.

Recap: Flexible demand can provide balancing value at lower cost than new supply. Poorly designed tariffs can burden users unable to shift; privacy and grievance safeguards matter.

Core Session 51: Energy poverty and just transition

```
ACCESS LADDER: connection -> hours/quality -> affordability
-> clean cooking -> productive use
TRANSITION: worker + community + local revenue + land + debt
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Energy poverty is deprivation of adequate, clean, reliable and affordable services; just transition manages concentrated adjustment losses.

Answer-grabbing line: Universal connection is an input, while welfare depends on quality, actual use and fair transition.

Must-write keywords: lifeline tariff; clean cooking; fuel stacking; reskilling; mine closure

Claim -> named evidence -> analysis -> qualification

- **Claim:** Universal connection is an input, while welfare depends on quality, actual use and fair transition.
- **Named evidence:** The 2018 GS-III PYQ links affordable, reliable, sustainable and modern energy to SDGs; coal-region assets create place-specific dependence.
- **Analysis:** Reliable service improves health, education and enterprise; transition support protects workers, communities and local finances.
- **Qualification:** Subsidised connection may not sustain use, and promised green jobs may not match location, wage or timing.

Evidence: The 2018 GS-III PYQ links affordable, reliable, sustainable and modern energy to SDGs; coal-region assets create place-specific dependence.

UPSC trap: Do not equate electrification with energy justice.

Mains use: Disaggregate households, workers, regions and time.

Recap: Universal connection is an input, while welfare depends on quality, actual use and fair transition. Subsidised connection may not sustain use, and promised green jobs may not match location, wage or timing.

Core Session 52: Transition finance and viability gaps

```
PROJECT CASH FLOW + POLICY CERTAINTY + COUNTERPARTY QUALITY
-> COST OF CAPITAL -> TARIFF
VGF buys justified viability, not all risk
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Transition finance mobilises long-tenure capital for grids, storage, clean supply, efficiency and adjustment.

Answer-grabbing line: For capital-heavy assets, financing cost can matter as much as equipment cost.

Must-write keywords: cost of capital; VGF; blended finance; payment security; green bond

Claim -> named evidence -> analysis -> qualification

- **Claim:** For capital-heavy assets, financing cost can matter as much as equipment cost.
- **Named evidence:** Economic Survey 2025-26 reports two BESS VGF schemes supporting about 43 GWh and an Rs 18,100 crore ACC battery PLI for 50 GWh, with 10 GWh earmarked for grid-scale storage.
- **Analysis:** VGF can enable socially valuable projects whose early commercial revenue is insufficient.
- **Qualification:** Support should not socialise avoidable execution risk or pay twice for one service.

Evidence: Economic Survey 2025-26 reports two BESS VGF schemes supporting about 43 GWh and an Rs 18,100 crore ACC battery PLI for 50 GWh, with 10 GWh earmarked for grid-scale storage.

UPSC trap: Do not equate outlay, award, financial closure and commissioning.

Mains use: Audit additionality, risk allocation, affordability and fiscal exposure.

Recap: For capital-heavy assets, financing cost can matter as much as equipment cost. Support should not socialise avoidable execution risk or pay twice for one service.

Core Session 53: Critical minerals and clean-energy supply chains

```
MINE -> CONCENTRATED PROCESSING -> CELL/MODULE/MAGNET
-> ASSET -> RECYCLING
NEW DEPENDENCE may replace hydrocarbon dependence
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Critical-mineral security concerns mining, processing concentration, technology, trade, safeguards and recycling.

Answer-grabbing line: A transition can reduce oil exposure while increasing mineral and component exposure.

Must-write keywords: critical minerals; processing; battery; magnet; recycling; circularity

Claim -> named evidence -> analysis -> qualification

- **Claim:** A transition can reduce oil exposure while increasing mineral and component exposure.
- **Named evidence:** Economic Survey 2025-26 identifies material intensity and critical-mineral access as transition constraints.
- **Analysis:** Diversified sourcing, domestic processing, R&D, substitution and recycling reduce concentration risk.
- **Qualification:** Domestic mining is not secure if processing, technology or environmental legitimacy is absent.

Evidence: Economic Survey 2025-26 identifies material intensity and critical-mineral access as transition constraints.

UPSC trap: Do not label every mineral critical without a stated context.

Mains use: Map the whole supply chain and qualify protection by cost.

Recap: A transition can reduce oil exposure while increasing mineral and component exposure. Domestic mining is not secure if processing, technology or environmental legitimacy is absent.

Core Session 54: Geopolitical shocks, routes and hedging

```
SUPPLY/ROUTE SHOCK -> landed fuel + freight/insurance
-> current account + rupee -> inflation -> fiscal/monetary trade-off
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: External shocks transmit through volumes, prices, routes, insurance and exchange rates.

Answer-grabbing line: Energy diplomacy should reduce expected disruption damage, not merely seek the cheapest spot cargo.

Must-write keywords: chokepoint; concentration; term contract; spot; hedge; current account

Claim -> named evidence -> analysis -> qualification

- **Claim:** Energy diplomacy should reduce expected disruption damage, not merely seek the cheapest spot cargo.
- **Named evidence:** PPAC dependence ratios and the 5.33 MMT Phase-I strategic reserve show why source, route, contract and stock diversification matter.
- **Analysis:** Term contracts reduce volume uncertainty; spot purchases add flexibility; hedges manage price.
- **Qualification:** A diversified supplier list can share one chokepoint; financial hedges cannot solve physical blockage.

Evidence: PPAC dependence ratios and the 5.33 MMT Phase-I strategic reserve show why source, route, contract and stock diversification matter.

UPSC trap: Do not call a futures hedge an emergency stock.

Mains use: Trace shock to balance of payments, rupee, inflation and response.

Recap: Energy diplomacy should reduce expected disruption damage, not merely seek the cheapest spot cargo. A diversified supplier list can share one chokepoint; financial hedges cannot solve physical blockage.

Core Session 55: Cyber and climate resilience

```
HAZARD -> ASSET/CONTROL-SYSTEM EXPOSURE -> OUTAGE CASCADE
REDUNDANCY + ISLANDING + BLACK START + HARDENING -> RECOVERY
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Resilience is the ability to absorb, adapt to and recover from shocks while maintaining critical service.

Answer-grabbing line: Efficiency and resilience can conflict when lean systems remove redundancy.

Must-write keywords: N-1; islanding; black start; cyber segmentation; climate stress

Claim -> named evidence -> analysis -> qualification

- **Claim:** Efficiency and resilience can conflict when lean systems remove redundancy.
- **Named evidence:** Grid digitisation and smart meters widen attack surfaces; heat, drought, flood and cyclone affect demand, lines, mines, hydro and cooling.
- **Analysis:** Redundancy, spare equipment, diversified siting and emergency protocols reduce recovery time.
- **Qualification:** No system is risk-free; hardening must prioritise critical loads and proportional cost.

Evidence: Grid digitisation and smart meters widen attack surfaces; heat, drought, flood and cyclone affect demand, lines, mines, hydro and cooling.

UPSC trap: Do not measure resilience only by annual outage averages.

Mains use: Use hazard, exposure, vulnerability, response and recovery.

Recap: Efficiency and resilience can conflict when lean systems remove redundancy. No system is risk-free; hardening must prioritise critical loads and proportional cost.

Core Session 56: Policy evaluation and cross-links

```
DELIVERED RELIABLE ENERGY + SYSTEM COST + FISCAL EXPOSURE
+ EMISSIONS/ECOLOGY + IMPORT RISK + DISTRIBUTION -> VERDICT
TOPIC 18 finance | TOPIC 25 climate
```

Visual first: The diagram fixes the mechanism and its boundary before the explanation.

Definition: Topic 31 owns energy-system economics; Topic 18 supplies infrastructure finance and Topic 25 supplies climate-economics depth.

Answer-grabbing line: The right metric is a dashboard because every single indicator can be gamed or misread.

Must-write keywords: unserved energy; system cost; fiscal exposure; emissions; import risk; distribution

Claim -> named evidence -> analysis -> qualification

- **Claim:** The right metric is a dashboard because every single indicator can be gamed or misread.
- **Named evidence:** Official reporting supplies capacity, generation, deficits, utility ratios, fuel dependence and programme status, while causal evaluation still needs counterfactual analysis.
- **Analysis:** A policy can improve one objective and worsen another: cheap tariffs can raise fiscal cost; domestic coal can lower import risk but raise pollution.
- **Qualification:** Correlation after a scheme is not attribution; national averages conceal household and state variation.

Evidence: Official reporting supplies capacity, generation, deficits, utility ratios, fuel dependence and programme status, while causal evaluation still needs counterfactual analysis.

UPSC trap: Do not rank policy by money spent or capacity sanctioned.

Mains use: Conclude on delivered service, full cost, fiscal durability, emissions and resilience.

Recap: The right metric is a dashboard because every single indicator can be gamed or misread. Correlation after a scheme is not attribution; national averages conceal household and state variation.

BASIC MCQS / REMEDIATION

Q1. Which statement correctly distinguishes primary and secondary energy?

A. Crude oil is primary, while petrol and electricity are secondary carriers. B. Electricity is always primary because it powers final use. C. Hydrogen is a geological primary resource in energy accounts. D. Final energy is measured before conversion losses.

Answer: A

Option explanations:

- **A is correct:** Crude exists before conversion; petrol and electricity are produced carriers.
- **B is incorrect:** Electricity normally results from conversion of a primary source.
- **C is incorrect:** Hydrogen must be produced and therefore acts as a carrier.
- **D is incorrect:** Final energy reaches the user after upstream conversion and network losses.

UPSC trap: Primary, secondary and final describe positions in a conversion chain.

Q2. Which classification is most accurate?

A. All biomass is non-commercial energy. B. Commercial/non-commercial and modern/traditional are separate axes. C. Every renewable source is non-commercial. D. Traditional fuel use proves absence of an electricity connection.

Answer: B

Option explanations:

- **A is incorrect:** Modern pellets and biogas may be traded commercially.
- **B is correct:** A fuel can be commercial yet traditional, or renewable yet modern.
- **C is incorrect:** Solar and wind electricity are commonly sold through markets and contracts.

- **D is incorrect:** Fuel stacking can persist after electrification because of affordability or reliability.

UPSC trap: Do not collapse market status into technology quality.

Q3. At 31 December 2025, what did the 51.93% figure denote?

A. Non-fossil share of total primary energy. B. Renewable share of final energy consumption. C. Non-fossil share of installed electricity capacity. D. Non-fossil share of April-December electricity generation.

Answer: C

Option explanations:

- **A is incorrect:** The Ministry table concerns installed power capacity, not primary energy.
- **B is incorrect:** No final-consumption denominator appears in that capacity table.
- **C is correct:** The annual report labels total non-fossil capacity as 51.93% of installed capacity.
- **D is incorrect:** April-December 2025 non-fossil generation was 30.41%, a separate flow.

UPSC trap: Capacity share and generation share use different denominators and periods.

Q4. A 100 MW battery with 400 MWh usable energy can ideally discharge at rated power for approximately:

A. One hour B. Two hours C. Three hours D. Four hours

Answer: D

Option explanations:

- **A is incorrect:** One hour would use only 100 MWh of the stated energy.
- **B is incorrect:** Two hours would use 200 MWh, half the stated store.
- **C is incorrect:** Three hours would use 300 MWh, below the stated store.
- **D is correct:** Energy divided by power gives four hours before efficiency and operating limits.

UPSC trap: Storage must state both MW and MWh.

Q5. Which statement about PLF is correct?

A. It measures utilisation of thermal capacity over a period, not thermal efficiency. B. It equals coal's share of total generation. C. It is the maximum demand met by a system. D. It measures billing and collection performance.

Answer: A

Option explanations:

- **A is correct:** PLF compares actual thermal output with output possible at rated capacity.
- **B is incorrect:** Generation share is an energy-mix ratio with another denominator.
- **C is incorrect:** Peak demand met is an instantaneous power statistic.
- **D is incorrect:** Billing and collection enter AT&C and revenue metrics.

UPSC trap: Low PLF has several possible causes; do not diagnose efficiency from it.

Q6. Which proposition about deficits is valid?

A. Zero peak deficit proves zero local outages. B. Peak deficit and energy deficit measure different dimensions of shortage. C. Energy deficit is measured only in MW. D. Peak deficit equals the ACS-ARR gap.

Answer: B

Option explanations:

- **A is incorrect:** National peak balance can coexist with local distribution failures.
- **B is correct:** Peak deficit concerns MW at the maximum interval; energy deficit concerns cumulative supply.
- **C is incorrect:** Energy deficit uses energy units or a percentage of requirement.
- **D is incorrect:** ACS-ARR is financial under-recovery in Rs/kWh.

UPSC trap: National adequacy is not identical to local service reliability.

Q7. Why is security-constrained dispatch broader than a simple merit order?

A. It ignores variable cost. B. It applies only to renewable generators. C. It includes congestion, reserves and technical operating limits. D. It fixes retail tariffs.

Answer: C

Option explanations:

- **A is incorrect:** Variable cost remains important within the eligible set.
- **B is incorrect:** Dispatch covers all scheduled resources under applicable rules.
- **C is correct:** Grid security, ramping, minimum load and congestion constrain pure cost ranking.
- **D is incorrect:** Retail tariff is determined by the relevant commission.

UPSC trap: Merit order is nested inside technical and security constraints.

Q8. Which institution-function pair is correct?

A. CEA - retail tariff adjudication B. CERC - household electrification execution C. SERC - national load despatch D. Grid Controller India - national and regional system operation

Answer: D

Option explanations:

- **A is incorrect:** CEA is the technical planning and standards authority.
- **B is incorrect:** CERC regulates specified inter-state and central matters.
- **C is incorrect:** SERC regulates state matters; SLDC performs state system operation.
- **D is correct:** Grid Controller India operates NLDC and regional load-despatch functions.

UPSC trap: Separate technical authority, regulator, network utility and operator.

Q9. What is the Electricity (Amendment) Bill, 2025 status at the cutoff used here?

A. A draft proposal, with no located enactment replacing the 2003 Act. B. A constitutional amendment already in force. C. A CERC tariff regulation. D. A state code automatically binding nationwide.

Answer: A

Option explanations:

- **A is correct:** The Ministry circulated a draft; no enactment was located by 10 September 2026.
- **B is incorrect:** This is not a constitutional-amendment measure.
- **C is incorrect:** A parliamentary Bill is not a commission regulation.
- **D is incorrect:** A state code cannot enact a central parliamentary Bill.

UPSC trap: Proposal, passage, assent and commencement are separate stages.

Q10. Which tariff statement is correct?

A. A low consumer tariff always proves low system cost. B. Explicit subsidy and cross-subsidy have different payers and incidence. C. Regulatory assets eliminate the underlying cost. D. Fixed charges vary only with each kWh.

Answer: B

Option explanations:

- **A is incorrect:** Low tariffs may conceal subsidy, debt or deferred maintenance.
- **B is correct:** Budget subsidy is paid by government; cross-subsidy is borne by another class.
- **C is incorrect:** A regulatory asset defers recovery and usually carries future cost.
- **D is incorrect:** Fixed charges recover costs not proportional to current energy use.

UPSC trap: Trace who pays now, later, through taxes or poor service.

Q11. Which description of open access is most defensible?

A. It provides free use of electricity networks. B. It abolishes universal-service obligations. C. It permits eligible users to choose supply while paying applicable network and system charges. D. It guarantees every purchase is renewable.

Answer: C

Option explanations:

- **A is incorrect:** Network use carries wheeling, transmission, loss and other lawful charges.
- **B is incorrect:** Distribution obligations continue under statute and licence.
- **C is correct:** Choice is combined with cost recovery for monopoly wires and system services.
- **D is incorrect:** The source depends on the transaction; green open access is a specified subset.

UPSC trap: Supplier choice does not mean costless wires.

Q12. Which statement about power exchanges is correct?

A. Day-ahead price is the retail tariff for all consumers. B. Exchanges replace every long-term PPA. C. Market coupling was universally complete before July 2025. D. They provide regulated short-term products while much procurement remains outside exchanges.

Answer: D

Option explanations:

- **A is incorrect:** Exchange price excludes many retail network, loss and policy components.
- **B is incorrect:** Long-term and bilateral contracts continue.
- **C is incorrect:** CERC's July 2025 direction initiated phased work rather than proving completion.
- **D is correct:** Day-ahead, real-time and term-ahead products form part of the procurement architecture.

UPSC trap: Do not generalise one short-term price to the whole sector.

Q13. What is the proper role of deviation settlement?

A. It prices departures from schedules to support discipline and balance. B. It is a long-term capacity contract. C. It replaces real-time system operation. D. It is a household efficiency subsidy.

Answer: A

Option explanations:

- **A is correct:** DSM compares actual injection or drawal with schedule and applies settlement.
- **B is incorrect:** Capacity contracts procure dependable capability over another horizon.
- **C is incorrect:** Operators still deploy reserves and manage congestion.
- **D is incorrect:** Demand-side efficiency is unrelated despite sharing the abbreviation DSM.

UPSC trap: Spell out DSM because it has two common meanings.

Q14. Which statement about ancillary services is correct?

A. They are measured only by annual MWh. B. They procure response and reserves needed for frequency and reliability. C. They are identical to cross-subsidy. D. Only coal plants can provide them.

Answer: B

Option explanations:

- **A is incorrect:** Response speed and availability matter in addition to energy.
- **B is correct:** Ancillary products support balancing and contingencies.
- **C is incorrect:** Cross-subsidy concerns tariff incidence between classes.
- **D is incorrect:** Hydro, storage, gas, demand response and eligible generators can contribute.

UPSC trap: Energy and reliability services are distinct products.

Q15. AT&C loss primarily combines:

A. Only transformer heat loss. B. Only unmetered farm supply. C. Technical, billing and collection losses. D. The difference between coal and gas prices.

Answer: C

Option explanations:

- **A is incorrect:** Physical loss is only one component.
- **B is incorrect:** Unmetered supply may affect estimation but does not exhaust the metric.
- **C is correct:** The metric links input energy to revenue actually collected.
- **D is incorrect:** Fuel-price comparison belongs to generation economics.

UPSC trap: AT&C is broader than theft and narrower than total financial loss.

Q16. Which Ministry of Power FY 2024-25 pair is correct?

A. AT&C 21.91%; ACS-ARR Rs 0.69/kWh B. AT&C 12%; ACS-ARR zero C. AT&C 5.28%; ACS-ARR Rs 15.04/kWh D. AT&C 15.04%; ACS-ARR Rs 0.06/kWh

Answer: D

Option explanations:

- **A is incorrect:** These are the FY 2020-21 starting values in the comparison.
- **B is incorrect:** These are RDSS targets, not the reported outcome pair.
- **C is incorrect:** The numbers and units are reversed; 5.28 crore refers to meter installations.
- **D is correct:** The annual report gives this FY 2024-25 national pair.

UPSC trap: Keep percentage loss and rupees-per-kWh gap in their units.

Q17. How should UDAY and RDSS be distinguished?

A. UDAY centred on a 2015 debt reset plus reforms; RDSS is a later conditional infrastructure and performance scheme. B. RDSS is a new name for UDAY debt bonds. C. UDAY began after RDSS. D. Both are sections of the Electricity Act.

Answer: A

Option explanations:

- **A is correct:** The schemes differ in date, instrument and principal mechanism.
- **B is incorrect:** RDSS works and smart metering are not the old debt takeover.
- **C is incorrect:** UDAY launched in November 2015; RDSS followed later.
- **D is incorrect:** They are policy schemes, not sections of the Act.

UPSC trap: A successor reform is not automatically a merger.

Q18. What does a zero ACS-ARR gap fail to prove by itself?

A. That a tariff order exists. B. That subsidy, dues and collections arrived on time and service improved. C. That average cost can be measured. D. That power purchase has a cost.

Answer: B

Option explanations:

- **A is incorrect:** A tariff order can exist regardless of the gap.
- **B is correct:** Accrual accounting can conceal cash timing, arrears and quality problems.
- **C is incorrect:** The indicator presupposes measured cost and revenue.
- **D is incorrect:** Power purchase remains a major ACS component.

UPSC trap: Pair booked gap with cash flow and service quality.

Q19. Which coal statement is correct?

A. Coking and non-coking coal have identical uses. B. Imports prove India has no domestic production. C. Production, dispatch, plant receipt and generation are distinct stages. D. A mine award guarantees immediate output.

Answer: C

Option explanations:

- **A is incorrect:** Coking coal is crucial for metallurgical use; non-coking serves power and others.
- **B is incorrect:** India produced 1,047.523 MT in FY 2024-25 while importing coal.
- **C is correct:** Quality and logistics create gaps between each stage.
- **D is incorrect:** Clearances, development and evacuation precede production.

UPSC trap: Follow the physical chain instead of treating one quantity as another.

Q20. Which pair accurately states FY 2025-26 coal imports?

A. Total 66.33 MT, all coking B. Total 180.04 MT, all non-coking C. Total 1,047.523 MT, all imports D. Total 246.37 MT: 66.33 coking and 180.04 non-coking

Answer: D

Option explanations:

- **A is incorrect:** 66.33 MT is only the coking component.
- **B is incorrect:** 180.04 MT is only the non-coking component.
- **C is incorrect:** 1,047.523 MT is FY 2024-25 domestic production.
- **D is correct:** The two components sum to the Ministry's reported total.

UPSC trap: Production and import flows require the right year and category.

Q21. Which statement about revised SHAKTI is accurate?

A. It is a power-sector coal-linkage framework revised with Cabinet approval on 7 May 2025. B. It regulates city-gas pipelines. C. It is a renewable-certificate market. D. It abolishes every coal auction.

Answer: A

Option explanations:

- **A is correct:** SHAKTI governs specified coal-allocation and linkage routes for power.
- **B is incorrect:** PNGRB regulates specified gas-network activities.
- **C is incorrect:** REC operates under CERC regulations.
- **D is incorrect:** The reform does not erase every auction or contract.

UPSC trap: Coal linkage, mine auction and e-auction are different.

Q22. Why can a coal plant become economically stranded?

A. Every plant must close at the same age. B. Lower utilisation, policy change or cheaper alternatives can erode cash flow before debt recovery. C. Sunk cost is always recoverable. D. Environmental controls have no cost.

Answer: B

Option explanations:

- **A is incorrect:** Retirement depends on economics, policy and system need.
- **B is correct:** Unexpected utilisation and revenue changes can destroy asset value.
- **C is incorrect:** Recovery depends on contracts and regulatory prudence.
- **D is incorrect:** Retrofits can require substantial capital and operating spending.

UPSC trap: Low PLF alone does not prove stranding.

Q23. Which petroleum-chain mapping is correct?

A. Upstream refining; downstream exploration B. Midstream retail pricing; upstream city gas C. Upstream exploration/production; midstream transport/storage; downstream refining/marketing D. PNGRB regulates only upstream production

Answer: C

Option explanations:

- **A is incorrect:** Refining is downstream and exploration is upstream.
- **B is incorrect:** Retail is downstream; city gas is a regulated network activity.
- **C is correct:** This is the standard functional chain.
- **D is incorrect:** PNGRB expressly excludes crude and gas production.

UPSC trap: Classify the activity before naming its regulator.

Q24. Which PPAC FY 2025-26 provisional inference is correct?

A. India produced more crude than products consumed. B. Gas imports were zero. C. Crude and gas had identical dependence. D. Crude dependence was 88.7%, while gas dependence was 50.1%.

Answer: D

Option explanations:

- **A is incorrect:** Crude output was 28.0 MMT and product consumption 241.6 MMT, although categories differ.
- **B is incorrect:** LNG imports were 34,427 MMSCM.
- **C is incorrect:** The ratios differ and use commodity-specific denominators.
- **D is correct:** This reproduces PPAC's ratios with the provisional-year caveat.

UPSC trap: Do not infer refinery weakness from crude dependence.

Q25. What is PNGRB's statutory perimeter?

A. Specified downstream/network petroleum and gas activities, excluding crude and gas production. B. All electricity tariffs. C. Coal-mine safety. D. Nuclear fuel regulation.

Answer: A

Option explanations:

- **A is correct:** This is the perimeter stated by PNGRB under its 2006 Act.
- **B is incorrect:** Electricity commissions regulate electricity tariffs.
- **C is incorrect:** Coal institutions perform mining functions.
- **D is incorrect:** Atomic-energy institutions govern nuclear fuel.

UPSC trap: The upstream-production exclusion is a frequent trap.

Q26. What does a strategic petroleum reserve do?

A. Permanently eliminates imports. B. Provides a finite physical buffer while wider diversification remains necessary. C. Hedges only the exchange rate. D. Guarantees indefinite retail-price stability.

Answer: B

Option explanations:

- **A is incorrect:** Finite inventory cannot replace continuing supply.
- **B is correct:** Stocks buy response time and complement source, route and contract diversity.
- **C is incorrect:** Financial instruments hedge price or currency; a reserve supplies crude.
- **D is incorrect:** Drawdown is limited and cannot neutralise every long shock.

UPSC trap: Capacity, fill level and days of cover are separate.

Q27. Which statement about renewable auctions is correct?

A. Tendered capacity is already generating. B. The winning bid is the complete retail tariff. C. A low bid still depends on finance, land, grid, commissioning and buyer payment. D. Auctions eliminate curtailment.

Answer: C

Option explanations:

- **A is incorrect:** Tender and commissioning are different stages.
- **B is incorrect:** Network, balancing and retail components remain.
- **C is correct:** These delivery conditions determine whether a bid becomes service.
- **D is incorrect:** Congestion and security constraints can still curtail output.

UPSC trap: Follow status from announcement to generation.

Q28. Which statement best explains curtailment?

A. It is always lack of sun or wind. B. It is storage discharge. C. It proves plant inefficiency. D. Available generation may be reduced because of congestion, security or demand constraints.

Answer: D

Option explanations:

- **A is incorrect:** Weather unavailability is not curtailment of available energy.
- **B is incorrect:** Storage discharge adds supply rather than withholding it.
- **C is incorrect:** A plant may be technically available when the grid cannot accept output.
- **D is correct:** This definition preserves the system-operator boundary.

UPSC trap: Variability, outage and curtailment are not synonyms.

Q29. Which GEC-II statement is correct?

A. It targets about 10,750 ckm and 27,500 MVA to integrate about 20 GW in seven states. B. It is a coal-mine scheme. C. Its design proves all lines are complete. D. It finances only rooftop panels.

Answer: A

Option explanations:

- **A is correct:** These are MNRE's official scheme-design figures.
- **B is incorrect:** The corridor concerns renewable-power transmission.
- **C is incorrect:** Targets do not establish completion.
- **D is incorrect:** The scheme builds intra-state transmission infrastructure.

UPSC trap: Circuit-km, MVA and GW measure different things.

Q30. Which statement about storage is correct?

A. Storage generates primary energy. B. Storage adds timing and reliability value but returns less energy than it absorbs. C. A MW rating fully states duration. D. Pumped storage has no ecological constraints.

Answer: B

Option explanations:

- **A is incorrect:** Storage shifts converted electricity; it does not create primary energy.
- **B is correct:** Round-trip loss is exchanged for timing and system service.
- **C is incorrect:** MWh and the MW/MWh ratio are required.
- **D is incorrect:** PSP depends on site, water, ecology and civil works.

UPSC trap: No storage claim is complete without power, energy and efficiency.

Q31. Which green-hydrogen status statement is accurate?

A. The 2030 target was achieved in 2023. B. Electrolyser capacity equals hydrogen output. C. The target and awards must be separated from commissioned production. D. Green hydrogen is mined.

Answer: C

Option explanations:

- **A is incorrect:** The mission sets a future annual-production target.
- **B is incorrect:** Electrolyser and hydrogen capacity use different units.
- **C is correct:** The Survey reports allocation and awards, not automatic operating output.
- **D is incorrect:** Hydrogen is produced as an energy carrier.

UPSC trap: Target, allocation, award and commissioning are separate.

Q32. Which comparison is correct?

A. REC and carbon credit are always the same. B. PAT ESCert certifies renewable generation. C. ESO is a crude-stock obligation. D. REC represents eligible renewable MWh; carbon credits and ESCerts have different bases.

Answer: D

Option explanations:

- **A is incorrect:** Renewable attributes and GHG units have different rules.
- **B is incorrect:** ESCert represents verified energy saving.
- **C is incorrect:** ESO concerns energy storage in the notified trajectory.
- **D is correct:** The instruments are not automatically fungible.

UPSC trap: One REC equals one eligible MWh, not one tonne CO2e.

PYQS AND ANSWER PRACTICE

Official-key discipline: Mains wording below is reproduced from the verified local official-paper route. These are learner model answers, not UPSC model answers. An objective answer letter is shown only after an exact authenticated question-paper/set-to-key mapping; no letter is inferred from a neutral route.

Verified Mains PYQ 1 - GS-III 2018 - 10 marks, 150 words

Question: "Access to affordable, reliable, sustainable and modern energy is the sine qua non to achieve Sustainable Development Goals (SDGs)." Comment on the progress made in India in this regard.

Model answer: India has moved from access scarcity toward service-quality and transition challenges. Electrification expanded network reach, while the Ministry of Power reports FY 2024-25 national energy and peak deficits of only 0.1% and 0.0%. Reliable electricity supports health facilities, digital education, irrigation and enterprise; clean cooking reduces indoor pollution and women's time burden. Yet connection is not continuous supply: affordability, voltage, outages, refill cost and productive use remain uneven. DISCOM weakness matters because a financially stressed last mile cannot maintain wires or procure power. Sustainability also requires efficiency, renewable integration, storage and cleaner fuels rather than only more capacity. Thus progress is substantial in physical access and aggregate adequacy, but SDG 7 demands an outcome test: affordable hours of quality power, clean cooking actually used, and resilient low-emission service for vulnerable households.

Why this earns marks: The directive is answered through claim -> named evidence -> analysis -> qualification, with unit, date, denominator and status preserved.

Verified Mains PYQ 2 - GS-III 2020 - 15 marks, 250 words

Question: Describe the benefits of deriving electric energy from sunlight in contrast to the conventional energy generation. What are the initiatives offered by our Government for this purpose?

Model answer: Solar electricity has no fuel purchase or combustion at generation, is modular and quick to deploy, and can serve utility, rooftop, pump and remote loads. It reduces marginal fuel-import exposure and local air pollution compared with fossil generation. Government instruments include solar parks and competitive procurement; PM Surya Ghar for rooftops; PM-KUSUM for farm applications; Green Energy Corridors; renewable obligations and RECs; domestic-manufacturing support; and storage procurement/VGF. Economic Survey 2025-26 reports 135.81 GW installed solar capacity by December 2025 and 8 GW rooftop capacity under PM Surya Ghar, but installed MW is not annual generation. Solar output is daytime-variable, needs land and transmission, and creates evening ramps, mineral and recycling issues. Solar's advantage is therefore strongest when capacity is integrated with grids, storage, demand response, domestic capability and ecological siting, and judged by reliable delivered energy rather than the winning bid.

Why this earns marks: The directive is answered through claim -> named evidence -> analysis -> qualification, with unit, date, denominator and status preserved.

Verified Mains PYQ 3 - GS-III 2021 - 10 marks, 150 words

Question: Explain the purpose of the Green Grid Initiative launched at World Leaders Summit of the COP26 UN Climate Change Conference in Glasgow in November 2021. When was this idea first floated in the International Solar Alliance?

Model answer: The Green Grids Initiative-One Sun One World One Grid seeks interconnected regional and global grids that move renewable electricity across time zones. A wider balancing area can use geographic diversity, share reserves, reduce curtailment and improve access to clean power. India first presented the transnational solar-grid idea at the International Solar Alliance's First Assembly in October 2018. Its economic logic is that sunlight available in one region can serve evening demand elsewhere through transmission. Yet cross-border lines do not automatically create trade. Common technical codes, transparent scheduling and settlement, finance, cyber security, sovereignty safeguards and mutual trust are required. It complements rather than replaces domestic corridors, storage and demand response. Interconnection can lower balancing cost and improve resilience only when market rules and political risks are managed.

Why this earns marks: The directive is answered through claim -> named evidence -> analysis -> qualification, with unit, date, denominator and status preserved.

Verified Mains PYQ 4 - GS-III 2022 - 15 marks, 250 words

Question: Do you think India will meet 50 percent of its energy needs from renewable energy by 2030? Justify your answer. How will the shift of subsidies from fossil fuels to renewables help achieve the above objective? Explain.

Model answer: The target must be stated correctly: India's COP26 commitment concerned about 50% of cumulative installed electric-power capacity from non-fossil sources by 2030, not half of total primary-energy consumption. India had reached 51.93% non-fossil installed capacity by 31 December 2025, while non-fossil generation was 30.41% in April-December 2025. Capacity therefore does not settle reliable energy contribution. Shifting support can improve relative prices by reducing general fossil consumption or producer support, financing renewable auctions, grids, storage, efficiency and targeted access, and internalising pollution and import risk. Abrupt withdrawal of lifeline electricity or clean-cooking support would be regressive; coal regions, workers and legacy debt need transition assistance. Renewable subsidies can also cause land, manufacturing or fiscal distortions. The appropriate shift is from general price suppression and unpriced externalities toward transparent, competitive support for generation, transmission, storage, demand flexibility and vulnerable users. Success should be measured by generation, curtailment, system cost, reliability and emissions, not capacity alone.

Why this earns marks: The directive is answered through claim -> named evidence -> analysis -> qualification, with unit, date, denominator and status preserved.

Verified Mains PYQ 5 - GS-III 2025 - 10 marks, 150 words

Question: How can India achieve energy independence through clean technology by 2047? How can biotechnology play a crucial role in this endeavour?

Model answer: Energy independence should mean lower strategic vulnerability, not autarky. India can reduce exposure through efficiency, electrification, diversified renewable and nuclear supply, stronger grids, storage, green hydrogen for hard-to-abate uses, domestic clean-technology manufacturing, critical-mineral partnerships and recycling. PPAC's provisional FY 2025-26 crude dependence of 88.7% shows why transport and industrial substitution matter, while reliability still requires firm capacity and flexible demand. Biotechnology can produce advanced biofuels, biogas and sustainable aviation-fuel feedstocks from residues, improve energy crops and microbial conversion, and support waste-to-energy pathways. Food-water-land competition, lifecycle emissions, biodiversity, logistics and cost must be audited. A resilient 2047 pathway combines demand reduction, clean carriers, domestic capability and diversified imports, judged by delivered cost, emissions and security rather than literal zero imports.

Why this earns marks: The directive is answered through claim -> named evidence -> analysis -> qualification, with unit, date, denominator and status preserved.

Objective PYQ official-key discipline

Year	Routed demand	Key treatment
2018	Solar power, silicon wafers and tariff regulation	Answer withheld: no authenticated local final key mapping used.
2019	PNGRB role and appellate route	Answer withheld: no authenticated local final key mapping used.
2022	Coal Controller's Organisation functions	Answer withheld: no authenticated local final key mapping used.
2023	Coal thermal plants, seawater and water stress	Answer withheld: no authenticated local final key mapping used.
2025	PNGRB regulated activities	Official Set-A key exists locally, but the full official option set is not reproduced here; no letter is inferred.
2025	Brazil/USA ethanol feedstock comparison	Official Set-A key exists locally, but the full official option set is not reproduced here; no letter is inferred.
2026	Green-hydrogen pathways and mission	Repository route labels the key provisional; no answer letter is recorded.

Rule: Concept teaching never manufactures an official option from a neutral routing summary.

ORIGINAL MAINS 1 - 10 MARKS - 150 words

Question: Distinguish installed capacity, generation, peak demand, energy supplied and final consumption. Why does the distinction matter for policy?

Model answer (138 native-body words):

Installed capacity is the rated power of generating assets, measured in MW or GW. Generation is electrical energy actually produced during a period, measured in MWh, GWh or BU. Peak demand is the highest instantaneous requirement; peak met records the maximum supplied at that interval. Energy supplied is the cumulative electricity delivered during a period. Final consumption is energy used by end-users after network and transformation losses and may include fuels beyond electricity. These distinctions change policy conclusions. At 31 December 2025, non-fossil sources were 51.93% of installed power capacity, but their April-December 2025 generation share was 30.41%. Storage also needs both MW and MWh. A credible evaluation must therefore attach unit, period, denominator and system boundary to every figure, then test whether capacity is available at peak, reaches consumers reliably and produces affordable useful service.

Examiner check: Directive fidelity, dated evidence, mechanism analysis and qualification are explicit.

ORIGINAL MAINS 2 - 10 MARKS - 150 words

Question: Why are DISCOM finances central to reliable electricity supply? Suggest a compact reform sequence.

Model answer (128 native-body words):

DISCOMs are the commercial hinge between generators and consumers. Inflow depends on accurate metering, billing, collection, timely state subsidy and government-dues payment; outflow includes power purchase, maintenance, employees, finance and capital works. Technical loss, theft, weak billing, delayed tariffs, unpaid subsidy and inflexible PPAs can create a cycle of borrowing, generator dues and poor service. Ministry of Power reports FY 2024-25 AT&C loss of 15.04% and ACS-ARR gap of Rs 0.06/kWh, but national averages do not prove every utility is healthy. Reform should sequence feeder and transformer accounting, reliable meters and grievance systems, targeted lifeline support, timely tariffs and subsidy payment, procurement review, loss-reduction investment and transparent regulation. RDSS should be judged by validated cash recovery, outage reduction and service, not meter installation alone.

Examiner check: Directive fidelity, dated evidence, mechanism analysis and qualification are explicit.

ORIGINAL MAINS 3 - 15 MARKS - 250 words

Question: Analyse the economics of integrating variable renewable energy into India's power system.

Model answer (175 native-body words):

Variable renewable energy has low fuel and short-run marginal cost, but output depends on weather and time. The problem shifts from procuring cheap plant-level energy to delivering reliable system service. At 31 December 2025, non-fossil sources formed 51.93% of installed capacity, while their April-December 2025 generation share was 30.41%; the different periods notwithstanding, this shows why nameplate capacity is not firm supply. Integration requires advance transmission, accurate forecasting, larger balancing areas, flexible hydro and thermal operation, storage, demand response, reserves and short-term markets. MNRE's GEC-II design aims at 10,750 circuit-km and 27,500 MVA to integrate about 20 GW. Economic Survey 2025-26 cites storage need of 336 GWh by 2029-30 and 411 GWh by 2031-32. Costs include curtailment, networks, balancing, land, minerals and recycling; benefits include avoided fuel, pollution and imports. Policy should procure portfolios and services rather than isolated megawatts: generation plus evacuation, duration-specific storage, time-of-day demand and resource adequacy. The verdict is pro-integration: success is lower unserved energy and total system cost, not the lowest bid or largest capacity addition.

Examiner check: Directive fidelity, dated evidence, mechanism analysis and qualification are explicit.

ORIGINAL MAINS 4 - 15 MARKS - 250 words

Question: Evaluate coal's changing role in India's energy security and transition.

Model answer (174 native-body words):

Coal remains central because existing plants, domestic mines and rail-linked infrastructure supply large volumes of dispatchable electricity and industrial fuel. The Ministry of Coal reports 1,047.523 MT production in FY 2024-25, yet FY 2025-26 imports were 246.37 MT, including 66.33 MT coking coal. Security depends on grade and delivered logistics, not aggregate production alone. Revised SHAKTI, approved 7 May 2025, addresses power-sector linkages, while commercial mine auctions concern another allocation route. Coal also imposes air, water, ash, land and carbon costs. As solar and wind expand, thermal plants may run fewer hours but need better ramping and reserve performance; lower PLF can raise per-unit fixed-cost recovery. Abrupt retirement can strand PPAs and debt and concentrate job, royalty and local-economy losses. Policy should prioritise efficient units, logistics, pollution compliance, flexible operation, transparent retirement criteria and closure funds while scaling grids, storage and demand response. A just transition needs worker protection, land restoration and regional diversification. Coal's role should decline through planned substitution of energy and reliability services, not a capacity-only pledge.

Examiner check: Directive fidelity, dated evidence, mechanism analysis and qualification are explicit.

ORIGINAL MAINS 5 - 20 MARKS - 250 words

Question: Assess India's oil and gas security architecture in the face of geopolitical and price shocks.

Model answer (186 native-body words):

India's vulnerability begins with high import dependence but extends to supplier concentration, shipping routes, currency, contracts, refining configuration and pricing. PPAC's FY 2025-26 provisional data report crude dependence of 88.7% and gas dependence of 50.1%. High crude dependence can coexist with strong refining and product exports, so the denominator matters. A shock raises the import bill, pressures the current account and rupee, transmits through fuel, freight and inputs, and creates a fiscal-monetary trade-off. The response requires layers. Diverse suppliers and routes reduce concentration; long contracts secure volume while spot cargoes retain flexibility; refinery complexity broadens crude choice; LNG terminals and common-carrier pipelines diversify gas access; efficiency and electrification reduce demand. The 5.33 MMT Phase-I strategic reserve is a finite buffer, while approved 6.5 MMT Phase-II capacity must not be counted as operating stock. Financial hedges manage price but not physical blockage. Targeted transfers are preferable to prolonged general price suppression. PNGRB's pipeline access can deepen markets, but affordability and anchor demand remain constraints. Security is diversified interdependence with buffers and substitution, measured by disruption recovery and delivered cost rather than zero imports.

Examiner check: Directive fidelity, dated evidence, mechanism analysis and qualification are explicit.

ORIGINAL MAINS 6 - 20 MARKS - 250 words

Question: Design a policy framework for a reliable, affordable and just Indian energy transition.

Model answer (189 native-body words):

A just transition must coordinate technology, networks, finance and distribution. First, plan from demand and reliability: distinguish MW from MWh, forecast peak and energy requirements, and procure dependable portfolios. Second, integrate renewables through corridors, storage, flexible hydro and thermal units, demand response, ancillary services and adequacy rules. Third, repair distribution through energy accounting, meters, timely tariffs and subsidy, targeted lifeline support and disciplined PPAs; FY 2024-25 national AT&C loss of 15.04% and ACS-ARR gap of Rs 0.06/kWh are progress indicators, not proof of universal health. Fourth, keep RPO/RCO, ESO, REC, PAT ESCerts and carbon credits distinct and prevent double counting. Fifth, reduce external vulnerability through efficiency, electrification, diversified oil and gas contracts, strategic stocks, domestic clean-tech capability, critical-mineral partnerships and recycling. Sixth, price pollution while financing reskilling, mine closure, regional diversification and clean cooking. VGF should buy measurable system services and not socialise avoidable project risk. Finally, publish a dashboard of unserved energy, outage quality, system cost, fiscal exposure, emissions, import concentration and distributional incidence. Success means dependable low-carbon service without shifting hidden debt, ecological damage or adjustment costs onto vulnerable consumers and regions.

Examiner check: Directive fidelity, dated evidence, mechanism analysis and qualification are explicit.

OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER

A. Welfare value and hidden cost

Social value includes consumer service, investment, reliability, innovation and security, less private system cost, fiscal cost, pollution, climate, ecological and transition costs. A low visible tariff can conceal debt or subsidy; a high tariff can conceal market power.

B. Natural monopoly and contestable layers

Generation and trading may support competition; transmission, system operation and distribution wires retain network-monopoly features. Retail choice therefore requires neutral metering and settlement plus fair recovery of wire and universal-service costs.

C. Utility death-spiral risk

High cross-subsidy can push industry toward captive/open access; the DISCOM loses profitable volume, raises charges and strengthens exit incentives. This is possible rather than automatic because demand growth and correctly designed network charges can moderate it.

D. Missing money and resource adequacy

Low-marginal-cost renewables can suppress energy prices while flexible capacity remains necessary in scarce hours. Compare adequacy obligations, ancillary markets, scarcity pricing, demand response and capacity contracts for cost, competition and gaming risk.

E. Renewable cannibalisation and duck curve

Simultaneous solar output lowers midday captured prices; sunset produces a steep net-load ramp. Storage, flexible hydro/thermal, shifted demand, hybrid procurement, geographic diversity and interconnection are substitutes and complements.

F. LCOE versus system value

LCOE is a plant-level lifetime cost. System value adds timing, location, dependable capacity, flexibility, transmission, curtailment and externalities. A higher-LCOE resource may still be valuable at a scarce hour or node.

G. Storage revenue stacking

Storage may earn from arbitrage, capacity, reserves, congestion relief, network deferral and backup. Avoid double payment for one capability and state duration, cycles, degradation and state-of-charge constraints.

H. Oil-shock macroeconomics

Oil-price rise -> import bill -> current-account and rupee pressure -> fuel/freight/input inflation -> lower real income -> monetary-fiscal trade-off. Tax cuts lose revenue; broad subsidies weaken targeting; reserves bridge only temporary disruption.

I. Carbon-price and power-market interaction

Carbon pricing raises fossil variable cost and changes merit order, but can raise tariffs and expose coal regions or trade-exposed industry. Revenue recycling, lifeline protection, competitiveness treatment and double-counting rules matter.

J. Scenario method

Stress-test high demand, rapid efficiency, delayed transmission/storage, fuel-route shock and extreme-weather/cyber outage. Compare unserved energy, consumer cost, fiscal exposure, emissions, imports and recovery time rather than one forecast.

CONSOLIDATED REGISTER NOTES

Energy-system spine and units

- Resource/import -> conversion -> network/storage -> distribution -> final and useful service.
- MW/GW = power; MWh/GWh/BU = energy; storage needs both.
- Capacity share != generation share != total-energy share; PLF is utilisation, not efficiency.
- Peak deficit != energy deficit; neither alone proves local quality.

Dated power dashboard

- 31 December 2025: total capacity 513,729.69 MW; fossil 48.07%; non-fossil 51.93%.
- April-December 2025: total generation 1,384.86 BU; fossil 69.59%; non-fossil 30.41%.
- FY 2024-25: generation 1,829.70 BU; energy deficit 0.1%; peak deficit 0.0%.
- August 2026 provisional: all-India peak demand and peak met 258,270 MW.

Law, institutions and markets

- Electricity Act, 2003: licensing, open access, CERC/SERC/APTEL, tariff and subsidy architecture.
- CEA = technical planning/data; CERC = inter-state/market; SERC = state/retail; Grid-India/SLDC = operation.
- IEGC 2023 in force from 1 October 2023. Draft Electricity (Amendment) Bill, 2025 remained unenacted at the cutoff.
- Schedule -> actual deviation -> DSM settlement -> ancillary response. Market coupling work was phased, not universally complete.

Tariff and DISCOM

- Tariff = power purchase + network/loss + O&M/finance + return.
- Explicit subsidy != cross-subsidy != regulatory asset.
- FY 2024-25: AT&C 15.04%; ACS-ARR gap Rs 0.06/kWh.
- UDAY (2015 debt reset) != RDSS (conditional works/metering; sunset 31 March 2028).

Coal, oil and gas

- FY 2024-25 coal production 1,047.523 MT.
- FY 2025-26 coal imports 246.37 MT: coking 66.33; non-coking 180.04.
- Petroleum upstream/midstream/downstream are distinct; PNGRB excludes crude/gas production.
- PPAC FY 2025-26 provisional: crude production 28.0 MMT, product consumption 241.6 MMT, crude dependence 88.7%; gas net production 34,326 MMSCM, LNG imports 34,427 MMSCM, consumption 68,753 MMSCM, dependence 50.1%.
- Phase-I SPR capacity 5.33 MMT; approved Phase-II 6.5 MMT is not operational stock by assertion.

Transition and integration

- 31 December 2025: solar 135.81 GW; wind 54.51 GW; hydro 50.91 GW; nuclear 8.78 GW.
- GEC-II design: 10,750 ckm; 27,500 MVA; about 20 GW; seven states; Rs 12,031.33 crore; 33% central assistance.
- Storage need cited by Economic Survey: 336 GWh by 2029-30 and 411 GWh by 2031-32; two VGF schemes support about 43 GWh.
- Green-hydrogen mission: at least 5 MMT/year by 2030; Survey reports 862,000 t/year allocated, 3,000 MW/year electrolyser awards and three port hubs. Award != output.
- RPO/RCO/ESO, REC, PAT ESCert and CCTS carbon credits are different instruments and units.

Security, justice and evaluation

- Security = availability + access + affordability + reliability + sustainability + resilience.
- Portfolio = efficiency + diverse domestic/import supply + routes/contracts + stocks + grid/storage + viable firms + cyber/climate recovery.
- Energy poverty tests connection, hours, quality, affordability, clean cooking and productive use.
- Just transition tests worker, community, local-revenue, land and debt exposure.
- Final dashboard: delivered reliability, total system cost, fiscal exposure, emissions/ecology, import concentration, distribution and recovery time.

Examiner traps

- Announced/tendered/awarded/under construction/commissioned/generated/supplied are different.
- Renewable != non-fossil; storage != generation; DSM-deviation != demand-side management.
- REC != carbon credit != ESCert; subsidy != cross-subsidy; AT&C != ACS-ARR.
- Crude dependence can coexist with product exports; reserve capacity does not prove fill or days of cover.

ASCII Master Flow Diagram

```
ASCII MASTER FLOW - PANEL 1/12: Energy-service boundary
PRIMARY RESOURCE / IMPORT -> CONVERSION -> NETWORK / STORAGE
-> DISTRIBUTION -> FINAL ENERGY -> USEFUL SERVICE -> DEVELOPMENT
TEST: delivered reliability and affordability, not capacity or connection alone

ASCII MASTER FLOW - PANEL 2/12: Accounting and units
PRIMARY != SECONDARY != FINAL != USEFUL ENERGY
MW/GW = power | MWh/GWh/BU = energy | storage needs MW + MWh
CAPACITY SHARE != GENERATION SHARE != TOTAL-ENERGY SHARE
31 Dec 2025 non-fossil capacity 51.93%; Apr-Dec 2025 generation 30.41%

ASCII MASTER FLOW - PANEL 3/12: Load and dispatch
LOAD CURVE -> BASE + INTERMEDIATE + PEAK REQUIREMENTS
ELIGIBLE CAPACITY -> security/technical constraints -> merit order -> dispatch
PLF/capacity factor = utilisation, not efficiency
FY25 energy deficit 0.1%; peak deficit 0.0%; local quality remains separate

ASCII MASTER FLOW - PANEL 4/12: Electricity chain
GENERATION -> CTU/STU -> GRID-INDIA/NLDC/RLDC/SLDC
-> DISCOM / OPEN ACCESS -> METER -> CONSUMER
REAL TIME: schedule -> deviation -> DSM -> ancillary response

ASCII MASTER FLOW - PANEL 5/12: Law and institutions
ELECTRICITY ACT, 2003 -> licensing + open access + tariff + consumer duties
CEA technical | CERC inter-state/market | SERC state/retail | APTEL appeal
DRAFT Electricity (Amendment) Bill, 2025 != enacted law at 10 Sep 2026

ASCII MASTER FLOW - PANEL 6/12: Tariff and DISCOM
TARIFF = power + network/loss + O&M/finance + return
SUBSIDY != CROSS-SUBSIDY != REGULATORY ASSET
AT&C FY25 15.04% | ACS-ARR FY25 Rs 0.06/kWh
UDAY debt reset (2015) != RDSS conditional reform (sunset 31 Mar 2028)

ASCII MASTER FLOW - PANEL 7/12: Coal system
MINE -> GRADE -> EVACUATION -> STOCK -> GENERATION -> ASH/CLOSURE
FY25 production 1,047.523 MT
FY26 imports 246.37 MT = coking 66.33 + non-coking 180.04
LINKAGE/SHAKTI != MINE AUCTION != E-AUCTION

ASCII MASTER FLOW - PANEL 8/12: Oil and gas
UPSTREAM -> MIDSTREAM -> DOWNSTREAM; PNGRB excludes production
PPAC FY26 provisional: crude dependence 88.7%; gas dependence 50.1%
LNG -> regas -> pipe -> CGD/fertiliser/power/industry
SPR Phase-I 5.33 MMT; approved Phase-II 6.5 MMT != operational stock

ASCII MASTER FLOW - PANEL 9/12: Renewable portfolio
SOLAR daytime variable | WIND seasonal | HYDRO flexible/hydrology bound
BIOMASS feedstock bound | NUCLEAR firm and institutionally bounded
AUCTION -> award -> finance -> grid -> commission -> generation
LOW BID != delivered tariff; NO FUEL COST != zero system cost

ASCII MASTER FLOW - PANEL 10/12: Integration and storage
FORECAST + TRANSMISSION/GEC + FLEXIBLE SUPPLY + DEMAND RESPONSE
+ BESS/PSP + RESERVES/MARKETS -> RELIABLE VARIABLE-RE SERVICE
CEA need: 336 GWh by 2029-30; 411 GWh by 2031-32
STATE MW + MWh + duration + cycles + efficiency

ASCII MASTER FLOW - PANEL 11/12: Instruments
RPO/RCO = renewable obligation | ESO = storage obligation
REC = 1 MWh renewable attribute | PAT ESCert = energy saving
CCTS credit = GHG performance; NO automatic fungibility
Green H2 target >=5 MMT/year by 2030; allocation != commissioning

ASCII MASTER FLOW - PANEL 12/12: Security and answer route
AVAILABILITY + ACCESS + AFFORDABILITY + RELIABILITY + SUSTAINABILITY + RESILIENCE
-> diversify source/route/contract + stocks + efficiency + finance + recovery
ANSWER: define/date/unit -> trace chain/institutions -> diagnose economics
-> test justice/environment/import risk -> verdict on delivered system cost
CROSS-LINKS: Topic 18 infrastructure | Topic 25 climate economics
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Answer spine

Define system, carrier, unit and date -> separate capacity/generation/peak/supply/consumption -> trace value chain and institution -> diagnose tariff, contract, DISCOM and fuel economics -> test security, grid integration, finance, justice and environment -> evaluate reliable delivered energy, total system cost, fiscal exposure, emissions and import risk -> conclude with a diversified, flexible and accountable transition.

Source register

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